

historical trough battery

<https://gemini.google.com/app/cd502d816727a98f>

User prompt: I'm thinking about making a replica of a trough battery. I'm wondering if there are any materials other than copper that would work as a primary cell with zinc anode.

Response: Several cathode materials can replace copper when paired with a zinc anode, ranging from direct historical substitutes to modern conductive materials.

Conductive & Catalytic Cathode Materials

- **Graphite / Porous Carbon:** The most practical and chemically stable alternative. Carbon acts as an inert electronic conductor for hydrogen evolution (or oxygen reduction if exposed to air). It yields an open-circuit potential around **1.2 V to 1.4 V** against zinc in dilute acid or neutral chloride/sulfate electrolytes.
- **Platinized Silver (The Smee Cell):** Historically developed by Alfred Smee in 1840 to replace copper in early trough and immersion batteries. The finely divided platinum black deposited onto silver drastically lowers the overpotential for hydrogen gas evolution, preventing hydrogen bubbles from clinging to the plate and resisting the rapid polarization typical of plain copper.
- **Silver Sheet:** Higher standard electrode potential than copper ($E^\circ = +0.80 \text{ V}$ vs. $+0.34 \text{ V}$ for copper), resulting in a cell voltage around **1.55 V** against zinc, though expensive for large plates.
- **Nickel or Stainless Steel (316L):** Functions similarly to copper in neutral salt or alkaline electrolytes with reasonable chemical resistance. However, in dilute sulfuric or hydrochloric acid, stainless steel may passivate or pit depending on the acid concentration, yielding a lower cell voltage (**~0.7 V to 0.9 V**).

Depolarized Metal Oxide Cathodes

In a simple trough with single-fluid electrolytes (like dilute sulfuric acid or brine), plain metal cathodes polarize rapidly due to insulating hydrogen gas formation ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$).

Incorporating an active depolarizer or metal oxide maintains a higher, steady current:

- **Manganese Dioxide (MnO_2) on Carbon:** Forming a composite plate or packing MnO_2 /carbon paste into porous cathode frames recreates the Leclanché primary chemistry, producing a steady **1.5 V** in ammonium chloride, zinc chloride, or sodium chloride electrolyte without significant gas evolution.
- **Lead Dioxide (PbO_2):** Used in early acid-based bichromate or acid-zinc cells. A PbO_2 plate paired with zinc in dilute sulfuric acid delivers an open-circuit potential over **2.4 V**, though it requires careful handling of lead and strong acid.

Cathode Material	Electrolyte	Approximate OCV	Primary Advantage / Behavior
Graphite / Carbon	Dilute H ₂ SO ₄ or NaCl / NH ₄ Cl	1.2 – 1.4 V	Zero corrosion; easy to source or fabricate into flat sheets.
Platinized Silver / Titanium	Dilute H ₂ SO ₄	~1.0 V	Fast hydrogen release; historical Smee battery design.
Silver	Dilute H ₂ SO ₄ or NaCl	~1.5 V	Higher voltage than copper; very low internal resistance.
C + MnO ₂ Matrix	15–20% NH ₄ Cl or ZnCl ₂	~1.5 V	Eliminates H ₂ polarization; high current capacity.
316 Stainless Steel	NaCl or Na ₂ SO ₄ (neutral)	0.7 – 0.9 V	Readily available rigid sheet stock for structural mockups.

User prompt: Would grafoil work? Not historically accurate but I have it

Response: **Yes, Grafoil will work chemically**, but you will need to handle its mechanical limitations.

Grafoil (flexible exfoliated graphite) is chemically inert, highly conductive in-plane, and completely resistant to acidic, neutral, or alkaline electrolytes. Paired with zinc, it functions as a carbon cathode for hydrogen evolution (or dissolved oxygen reduction), producing an open-circuit voltage around **1.2 V to 1.4 V**.

Practical Considerations for a Trough Setup

- **Mechanical Rigidity:** Standard trough cells (like the Cruickshank design) rely on stiff plates slotted into partition grooves to separate compartments. Grafoil is soft and will flop or tear under the weight of sloshing electrolyte. It works best if laminated or mechanically clamped to a rigid, non-conductive backing (e.g., acrylic, 3D-printed frames, or thin plastic sheet).
- **Bipolar Partitioning (If Building a Series Trough):** In a traditional trough battery, the partition itself is a bipolar plate (zinc on one side, copper on the other). If you want the plate to act as the watertight divider between cells:
 - Grafoil is slightly permeable through its thickness if uncompressed, but dense sheets can work.
 - Joining zinc directly to Grafoil requires mechanical clamping or a conductive contact (soldering to graphite is not possible without specialized plating). Clamping a sheet of Grafoil flat against a zinc plate creates a simple, low-resistance bipolar divider.
- **Electrical Termination:** Since you cannot solder directly to Grafoil, use wide-area mechanical clamps (like plastic clips over a metal contact strip above the waterline) to prevent

tearing the foil while drawing current.

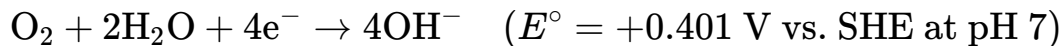
- **Surface Area & Polarization:** Grafoil's microscopic surface roughness actually gives it an advantage over polished copper sheet, providing slightly more active surface area for hydrogen bubble nucleation and release.

User prompt: Does the copper sheet evolve hydrogen in NaCl vs zinc?

Response: In a neutral NaCl solution, copper primarily reduces dissolved oxygen rather than evolving hydrogen under open-circuit and light-load conditions.

Hydrogen evolution can occur, but only under specific high-current or deoxygenated conditions.

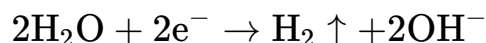
1. Open-Circuit and Normal Discharge (Oxygen Reduction) In aerated neutral saltwater (pH $\approx$ 7), the thermodynamically and kinetically favored cathodic reaction on copper is the **Oxygen Reduction Reaction (ORR)**:



Paired with zinc oxidation at the anode ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$, $E^\circ = -0.763 \text{ V}$), this yields a theoretical cell voltage of roughly **1.16 V**. Under standard conditions, no gas bubbles form at the copper plate; current is sustained entirely by dissolved oxygen diffusing to the copper surface.

2. Heavy Load / Short-Circuit (Hydrogen Evolution) If the cell is short-circuited or placed under a heavy electrical load, the demand for electrons quickly outpaces the rate of dissolved oxygen diffusion through the electrolyte (the mass-transport limit).

When the potential of the copper electrode polarizes down below the hydrogen equilibrium potential at pH 7 ($E = -0.414 \text{ V vs. SHE}$), plus copper's hydrogen overpotential, water reduction takes over:



Under these conditions, you will observe tiny hydrogen bubbles nucleating and clinging to the copper plate.

Why NaCl Differs from Acid Electrolytes

- **In Dilute Acid (e.g., H_2SO_4):** High concentration of hydronium ions (H^+) means hydrogen evolution happens immediately and vigorously upon load ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$).
- **In Neutral NaCl:** The extremely low H^+ concentration (10^{-7} M) makes proton reduction negligible. Current relies on dissolved O_2 (no bubbling) until the cathode potential is pulled low enough to directly split neutral water molecules.

User prompt: Would copper mesh work in a zn cu cell?

Response: **Yes, copper mesh will work—and in many ways, it will outperform a solid copper sheet.**

Using a mesh electrode offers distinct electrochemical advantages depending on the electrolyte and load conditions:

Key Advantages

- **Higher Active Surface Area:** Fine mesh significantly increases the true electrochemically active surface area (ECSA) per unit of frontal area compared to a flat sheet.
- **Better Oxygen Access in Neutral Brine (NaCl):** Since the primary cathodic reaction in neutral salt solution is dissolved oxygen reduction ($\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$), mass transport of O_2 to the electrode is the main bottleneck. The openings in a mesh allow convection and faster electrolyte circulation, raising the limiting current density before the cell polarizes.
- **Improved Bubble Detachment in Acid (H_2SO_4):** If operated in dilute acid or under heavy loads where hydrogen evolution occurs ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$), gas bubbles escape more easily through the apertures rather than forming a continuous insulating film across a flat plate.

Practical Trade-offs in a Trough Setup

- **Structural Rigidity:** Fine copper wire cloth or woven mesh will sag in liquid. It needs to be folded, stretched across a non-conductive carrier frame, or made from a thicker expanded copper mesh to stay flat and avoid shorting against the zinc anode.
- **Bipolar Partitioning:** In a classic Cruickshank-style trough where each divider is a solid, fluid-tight bipolar plate (zinc on one side, copper on the other), mesh cannot serve as the watertight partition wall on its own. You would need to mount the mesh against a solid backing plate or use a single-fluid immersion trough design with separate, suspended electrodes.

User prompt: More surface area than a solid foil? Maybe a double or triple layer.

Response: **Yes, fine copper mesh generally provides significantly more true electrochemically active surface area than flat foil, and multi-layering can push that area even higher.**

A quick geometric comparison clarifies the surface area multiplier and how multiple layers perform:

Flat Foil vs. Woven Mesh Area

For a flat square sheet of side length L submerged on both sides:

$$\text{Area}_{\text{foil}} = 2L^2$$

For a square wire mesh of side length L , wire diameter d , and mesh count n (wires per unit length):

- The total wire length in one direction is $nL \times L = nL^2$.
- Accounting for both warp and weft wires: Total wire length = $2nL^2$.
- Approximating each wire as a cylinder (surface area per length = πd):

$$\text{Area}_{\text{mesh}} \approx 2\pi ndL^2$$

Taking the ratio of mesh surface area to flat foil area:

$$\text{Ratio} = \frac{2\pi ndL^2}{2L^2} = \pi nd$$

- **Open Area vs. Wire Area:** The product $n \cdot d$ is the fraction of the linear dimension blocked by wire. For typical plain weave meshes, $n \cdot d$ is often between 0.35 and 0.60.
- **Area Factor:** Multiplying by π gives an area ratio typically between $1.1\times$ and $1.9\times$ compared to a smooth flat foil of the same footprint.
- **Micro-Roughness:** Drawn wire also possesses micro-scale surface roughness and curvature, promoting better fluid contact and bubble release.

Layering Multiple Mesh Sheets (Double/Triple Layer)

Stacking 2 or 3 layers multiplies the solid surface area by $2\times$ or $3\times$, but **electrochemical utility depends heavily on mass transport and spacing**:

- **In Neutral NaCl (Oxygen Diffusion-Limited):**
 - If layers are tightly compressed together, the inner surfaces become starved of dissolved O_2 , causing a "shadowing" effect where the interior layers contribute very little current.
 - Leaving a slight gap (even 0.5–1 mm) or using a looser open mesh allows natural buoyant convection of electrolyte, ensuring all layers see fresh dissolved oxygen.
- **In Acid (Hydrogen Evolution):**
 - Extra layers lower the true local current density ($i = I/A$), which directly reduces the **activation overpotential** required to evolve H_2 .
 - The open apertures provide clear release paths for gas bubbles to escape upwards instead of getting trapped between layers.

To maximize current in a static bath, loosely tacking 2–3 layers together with spot crimps (or folding an expanded mesh into an accordion) provides high surface area while keeping the structure open

to fluid flow.

User prompt: I assume these things will work better if the gap between the zinc and the copper on the next bipolar plate is as small as possible?

Response: **Up to a point, yes—reducing the inter-electrode gap lowers the internal electrolyte resistance, but going too narrow introduces severe mass-transport and clogging bottlenecks.**

In electrochemical cells, cell performance is a balance between **ohmic (IR) drop** and **fluid/mass transport dynamics**.

The Primary Benefit: Lower Ohmic Resistance (IR Drop)

The bulk resistance of the electrolyte between the zinc and copper plates is:

$$R_{\text{electrolyte}} = \frac{\rho \cdot d}{A} = \frac{d}{\kappa \cdot A}$$

- d = distance between plates (gap)
- κ = ionic conductivity of the electrolyte
- A = plate area

Halving the distance d cuts the electrolyte's contribution to internal resistance directly in half, giving a flatter discharge curve and higher usable current under load.

The Trade-offs & Practical Lower Limits

If the gap is made too narrow (e.g., less than 2–3 mm in a static, unpumped trough), several negative effects quickly overcome the ohmic advantage:

- **Precipitation & Clogging (In Neutral NaCl):**
 - Zinc oxidizes to Zn^{2+} , while the cathode generates hydroxide ions (OH^-).
 - In neutral brine without strong flow, these ions meet in the bulk gap and precipitate as insoluble **zinc hydroxide / oxychloride** ($\text{Zn}(\text{OH})_2$ / $\text{Zn}_5(\text{OH})_8\text{Cl}_2 \cdot \text{H}_2\text{O}$).
 - In an ultra-narrow gap, this white sludge cannot settle out to the bottom of the trough, bridging the gap, choking ion transport, and causing localized high-resistance dead zones.
- **Bubble Trapping (In Acid / High-Current Conditions):**
 - If hydrogen gas bubbles evolve on the cathode, they need room to rise and detach via buoyancy.

- In very tight gaps, capillary forces hold bubbles between the plates. This **gas void fraction** displaces conductive electrolyte with insulating gas, drastically *increasing* the effective electrical resistance.
- **Oxygen Diffusion Stagnation (In Neutral Brine):**
 - Since the cathode depends on dissolved O_2 , it relies on density-driven natural convection to bring fresh electrolyte into the space between the plates. A tight gap chokes this convective flow.
- **Mechanical Shorting:**
 - Zinc anodes corrode unevenly and can shed small metallic flakes, while thin cathode foils or mesh can warp. A minimal gap increases the risk of direct physical short circuits.

Practical Rule of Thumb

- **Static (Unstirred) Trough:** A spacing of **4 mm to 8 mm** (roughly 3/16" to 5/16") typically hits the sweet spot—narrow enough to keep IR drop reasonable in conductive brine/acid, but wide enough for natural convection, bubble release, and sediment clearance.
 - **Forced Circulation:** If electrolyte is actively pumped or stirred through the trough, the gap can safely drop down to **1 mm to 2 mm**.
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User prompt: What limits the capacity of a zinc copper cell?

Response: In a primary zinc-copper cell, capacity (total milliamp-hours, **mAh**, delivered before the voltage collapses) is almost never limited by the initial mass of metallic zinc. Instead, it is governed by **electrolyte volume, chemical precipitation, self-discharge, and cathode mass-transport limits**.

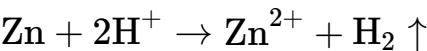
Primary Limiting Factors

- **1. Electrolyte Depletion and Solution Saturation**
 - **Neutral NaCl / Brine:** As zinc dissolves ($Zn \rightarrow Zn^{2+} + 2e^-$) and the cathode generates hydroxide (OH^-), insoluble zinc hydroxide and zinc oxychlorides ($Zn_5(OH)_8Cl_2 \cdot H_2O$) precipitate out. Once the local electrolyte is depleted of available counter-ions or choked with precipitate, ionic conductivity plummets and the cell dies long before the zinc plate is consumed.
 - **Dilute Acid (H_2SO_4):** Capacity is directly limited by the total moles of hydronium ions (H^+). For every mole of zinc oxidized, 2 moles of H^+ are consumed to evolve hydrogen gas. When the acid is neutralized to a neutral zinc sulfate solution, the cell voltage drops sharply.
- **2. Zinc Passivation (Fouling of the Anode)**

- In static electrolytes, dissolved Zn^{2+} ions accumulate directly at the anode surface faster than they can diffuse away.
- The local pH rises, forming an adherent, non-conductive crust of ZnO / $\text{Zn}(\text{OH})_2$ across the anode face. This passivation layer blocks further electrochemical dissolution of the underlying metal, causing an apparent "dead cell" even with substantial zinc remaining.

• **3. Open-Circuit Parasitic Self-Discharge**

- Zinc is thermodynamically unstable in aqueous electrolytes. Even with no external load connected, local galvanic micro-cells form on the zinc surface (spurred by trace iron, lead, or copper impurities in the sheet), directly evolving hydrogen gas:



- In acid, this corrosion can consume the entire zinc anode in hours to days without delivering useful work to the external circuit. *(Historically, this was mitigated by amalgamating the zinc with mercury).*

• **4. Cathode Polarization & Dissolved Oxygen Starvation (in Neutral Salt)**

- If the cell relies on the oxygen reduction reaction ($\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$), it delivers continuous current only as fast as atmospheric O_2 can dissolve and diffuse to the copper plate.
- If discharged at a rate faster than ambient replenishment, the cathode polarizes into deep water reduction, drastically lowering cell operating voltage and usable power output.

Theoretical vs. Practical Limits

Factor	Theoretical Limit	Practical Trough Limit
Zinc Metal Mass	820 mAh / gram of Zn	Rarely achieved (typically < 20–30% utilization before fouling)
Dilute Acid (1 M H_2SO_4)	≈ 53.6 Ah / liter	Governs runtime entirely unless fresh acid is added
Neutral Brine	Limited by O_2 solubility (≈ 8 mg/L)	Rate-limited by ambient diffusion; choked by $\text{Zn}(\text{OH})_2$ sludge

In practice, a trough cell running on a single charge of static electrolyte runs out of usable liquid capacity (acid consumed or brine passivated by sludge) long before the metal plates are physically consumed.

User prompt: How stiff is a 1.2 mm or 1 mm sheet of copper?

Response: A 1.0 mm to 1.2 mm sheet of copper is rigid and self-supporting at small-to-medium spans (like individual battery plates), but can still be bent cleanly by hand over a sharp edge or using hand tools.

Tactile Feel & Rigidity

- **1.0 mm (~18–19 AWG / ~2.9 lb/sq ft):** Feels substantial—similar in thickness to a standard credit card (~0.76 mm) or a US dime (1.35 mm). It will not flop, droop, or sag under its own weight. You cannot easily flex a 4" × 4" plate with your fingers alone, though a long strip can be bent with firm two-handed leverage.
- **1.2 mm (~16–17 AWG / ~3.5 lb/sq ft):** Visibly stiffer. Because flexural rigidity scales with the **cube of thickness** ($D \propto t^3$), a 1.2 mm sheet is roughly $1.73\times$ **stiffer** against bending than a 1.0 mm sheet of the same alloy and temper:

$$\left(\frac{1.2}{1.0}\right)^3 = 1.728$$

The Role of Temper (Hardness)

How "stiff" the sheet feels in practice depends heavily on whether it is annealed (soft) or cold-rolled (hard):

- **Soft / Annealed (O60):** Flexes easily under deliberate hand pressure. Once bent, it takes a permanent set with virtually zero springback. It stays flat if undisturbed, but drops into grooves or clamps without fighting you.
- **Half-Hard (H02) or Hard (H04):** Significantly higher yield strength. It resists incidental dings and bending, springs back elastically against light loads, and feels noticeably more rigid when handled.

Suitability for a Trough Battery Divider

For a DIY trough cell:

- **Slotting into Grooves:** 1.0 mm to 1.2 mm is a sweet spot. It is rigid enough to slide into slotted side-wall guides and compress against a gasket or beeswax/pitch seal without buckling under fluid pressure.
- **Cutting & Working:** It can be cut with heavy hand shears, a throatless shear, or a metal-cutting blade on a saw. It bends crisply in a vise or sheet metal brake without tearing.

Sources

1. [K-Factor for Copper: C110 Bend Allowance & Chart - Sheet Metal Fabrication](#)

User prompt: In copper, what's the difference between HO2 temper and soft temper?

Response: The difference between **Soft (Annealed / O-temper)** and **H02 (Half-Hard)** copper comes down to **cold working (work hardening)** and heat treatment, which drastically alters yield strength, stiffness to permanent bending, and formability while leaving electrical conductivity virtually unchanged.

Key Mechanical Differences (C11000 / Electrolytic Tough Pitch Copper)

Property	Soft / Annealed (O60)	Half-Hard (H02)	Impact on Workshop Use
Yield Strength	~70 MPa (10 ksi)	~250–290 MPa (36–42 ksi)	~3.5–4× higher in H02; resists permanent bending under hand pressure.
Tensile Strength	~220 MPa (32 ksi)	~290–340 MPa (42–49 ksi)	H02 supports higher ultimate mechanical loads before tearing.
Elongation at Break	40% – 50%	10% – 20%	Soft copper stretches significantly before failing; ideal for deep drawing or flanging.
Hardness (Rockwell F)	~40–45 HRF	~75–85 HRF	H02 resists surface scratches, indentation, and edge burring.
Springback	Virtually zero	Moderate	H02 "fights back" slightly when bent; soft stays exactly where you push it.
Electrical Conductivity	100–101% IACS	~98–100% IACS	Negligible difference (~1–2% drop due to crystal dislocations in H02).

How They Behave in the Shop

- **Soft (Annealed):** Dead-soft and pliable. If you bend a sheet over a 90° corner, it forms with very little force and retains the exact profile without springing back. However, thin sheets will easily take accidental dings, warp when handled roughly, and may "potato-chip" if you try to cut them with hand snips.
- **H02 (Half-Hard):** Produced by cold rolling the copper to reduce its thickness by roughly 20% after annealing. It feels springy and rigid. It resists accidental deformation, slides cleanly into tight partition slots without bowing, and maintains flat faces under moderate pressure. It can still be bent to 90° over a radius, but requires more bending force and exhibits slight springback.

Annealing & Work Hardening Copper switches between these states readily. Bending, hammering, or rolling soft copper work-hardens it toward H02/H04. Heating H02 copper with a torch to a dull red (~500–600°C) and quenching in water instantly returns it to dead-soft temper.

User prompt: What about 101 alloy vs 110

Response: The primary difference between **Alloy 101 (OFE - Oxygen-Free Electronic)** and **Alloy 110 (ETP - Electrolytic Tough Pitch)** is **purity and oxygen content**, which impacts high-temperature processing and hydrogen embrittlement rather than room-temperature electrical performance.

Key Chemical & Physical Differences

Property	C11000 (110 ETP)	C10100 (101 OFE)	Practical Impact
Purity	99.90% Cu (min)	99.99% Cu (min)	101 is the highest-purity standard commercial copper.
Oxygen Content	200 to 400 ppm (0.02–0.04%)	< 5 ppm (0.0005%)	110 contains microscopic cuprous oxide (Cu ₂ O) particles; 101 is truly oxygen-free.
Electrical Conductivity	100% to 101.5% IACS	101% to 102% IACS	Negligible difference for power and electrochemical circuits.
Hydrogen Embrittlement Resistance	Poor (prone to cracking if heated in reducing gas)	Immune	101 can be torch-brazed or annealed in hydrogen/reducing flames safely.
Cost & Availability	Standard commercial grade; low cost, universally stocked.	Premium specialty grade; significantly more expensive and harder to find.	110 represents >90% of all standard copper busbars, sheet, and pipe.

When the Difference Matters

- 1. Hydrogen Embrittlement (The Main Engineering Factor):
 - In **110 ETP**, dissolved oxygen forms tiny Cu₂O inclusions at grain boundaries. If 110 is heated above ~370°C (700°F) in an atmosphere containing hydrogen or reducing gases (such as a rich oxy-fuel torch flame), hydrogen diffuses into the metal, reacts with the oxide to form high-pressure steam (Cu₂O + H₂ → 2Cu + H₂O), and causes microscopic grain-boundary cracking and brittleness.
 - 101 OFE** has virtually no oxygen, making it completely immune to hydrogen embrittlement during high-temperature brazing, welding, or vacuum-furnace operations.
- 2. Ultra-High Vacuum (UHV) & Specialized Electronics:

- 101 OFE is specified for particle accelerators, semiconductor tools, and vacuum tubes because it resists outgassing when baked under hard vacuum.

Practical Verdict for Battery Cells and Shop Work

For battery plates, electrochemical cells, busbars, and general DIY fabrication:

- **C110 (ETP)** is the standard, economical choice. Its electrical conductivity is essentially identical to 101, and it solders, cuts, and bends perfectly at room temperature.
- **C101 (OFE)** offers no functional electrochemical or electrical advantage in an ambient aqueous cell, and is only worth sourcing if you plan on high-temperature furnace brazing in a reducing atmosphere or building ultra-high vacuum equipment.

User prompt: Hmm, so with the materials that I have I really need something stiff that can be sealed around the edges. And be able to fasten the zinc and copper to it. Doesn't have to be conductive but that would be a bonus. If I printed a trough with solid dividers and left slots on either side that the foil could slide into that would work. I have some nickel strips that could be bent over the top of the dividers to connect the metal on both sides.

Response: That is a solid, clean mechanical approach. Using solid printed partition walls and running the electrical connection **over the top** bypasses the biggest headache of historic Cruickshank troughs: trying to make a watertight liquid seal around a metal sheet passing through the divider.

Design & Mechanical Details to Consider

- **Watertight Solid Dividers:**
 - To guarantee cell-to-cell isolation with 3D-printed walls, use at least **4–6 solid perimeter perimeters (walls)** and ~100% infill (or concentric top/bottom layers) for the partition bulkheads. This eliminates microscopic pinhole porosity between extrusion paths that could cause internal ionic "short circuits" or electrolyte weeping across cells.
 - A slight draft angle or radius at the base of the slot channels will prevent stress cracking when sliding foils in.
- **Top-Bridge Contact Geometry (The Nickel Strap):**
 - Nickel strip (like 0.15 mm or 0.2 mm battery tab strip) is ductile and bends easily into a tight "U" or "saddle" clip over the divider rim.
 - **Dry vs. Wet Termination:** Keep the mechanical junction between the nickel strap and the foil **above the electrolyte waterline**. If the copper-nickel or zinc-nickel physical interface gets submerged in brine/acid, local galvanic micro-cells will form right at the joint, causing premature corrosion or contact resistance spikes.
 - **Clamping Pressure:** A simple 3D-printed friction-fit clip or a printed screw clamp slipping over the top edge will hold the nickel strap firmly pressed against the copper foil on one side and the zinc sheet on the other.

- **Slotted Retention Rails:**

- Integrate vertical channels (e.g., 1.5 mm wide × 1.5 mm deep) along the inside face of each cell wall.
- This holds the thin foils/Grafoil flat and maintains an exact, uniform gap (e.g., 5–6 mm) across the full face without bowing toward each other.
- Leaving a 5–10 mm clearance zone at the *bottom* of the slots creates a collection sump where precipitate ($\text{Zn}(\text{OH})_2$ sludge or zinc debris) can settle without shorting the plates or clogging the active surface.

- **Material Selection:**

- **PETG** or **ABS/ASA** are ideal for electrolyte troughs—they resist dilute acids, salts, and zinc chloride without swelling or hydrolyzing. Standard PLA works fine for temporary neutral NaCl testing, but can slowly embrittle or creep over time under sustained liquid exposure.

User prompt: Were sumps a part of historical trough cells?

Response: **No, early Cruickshank-style trough cells did not have sumps.**

In William Cruickshank's original 1800/1802 design, the metal plates themselves served as the physical bulkheads dividing the long wooden box into individual liquid compartments.

How Historical Trough Cells Were Built

- **Plates Reached the Floor:** Square bimetallic pairs (a zinc plate and copper plate soldered back-to-back) were slid into vertical grooves cut along the wood sides and bottom, then sealed directly into the wood using a resinous cement (typically a mixture of pitch, rosin, and beeswax or shellac).
- **No Under-Plate Clearance:** Because the plates had to create a watertight seal from top to bottom, there was no continuous cavity or clearance underneath the electrodes. Sediment accumulated directly on the floor between the plates.

The Evolution of the "Plunge" and Suspended Cell

The lack of a sump, combined with rapid plate fouling and the inability to stop parasitic zinc corrosion when not in use, quickly proved to be the trough's biggest operational flaw. This led directly to redesigns:

- **The Babington / Wollaston "Plunge" Battery (c. 1815):** William Babington and William Hyde Wollaston replaced the bimetallic partition design. They used porcelain or earthenware troughs with **fixed, solid dividers**, suspending the zinc and copper plates from a wooden beam above the trough.
- **Accidental Sumps:** This allowed the experimenter to:

1. Lift the plates out entirely when the battery was idle to stop acid corrosion.
2. Leave open space beneath the suspended plates where sediment and falling sludge could settle without clogging the active faces or causing bridging shorts.

By the mid-to-late 19th century—culminating in flooded lead-acid designs (like Gaston Planté and Camille Faure's cells)—dedicated **sediment spaces / mud ribs** at the bottom of the container became standard engineering practice to keep flaked oxide debris from bridging the plates.

Sources

1. [All you need to know about batteries - Borregaard](#)
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User prompt: What are pitch and rosin? I know what beeswax is

Response: **Rosin** and **pitch** are traditional resinous binders derived from pine trees (or fossil fuels), historically used as the standard waterproof, acid-resistant sealants and adhesives before modern synthetic polymers and silicones existed.

What is Rosin (Colophony)?

Rosin is the solid, amber-colored natural resin left behind after distilling liquid pine oleoresin to extract volatile **turpentine** (spirits of turpentine).

- **Properties:** It is hard, brittle, translucent, and easily crushed into a fine powder. It melts at roughly **70°C to 100°C (160°F to 212°F)** into a sticky liquid.
- **Chemical Nature:** Primarily composed of **abietic acid** and related diterpene resin acids. It is insoluble in water and highly resistant to acids, but dissolves in alcohols, acetone, and hot oils.
- **Familiar Modern Uses:** The solid amber cake used on violin bows, the grip powder used by baseball pitchers and gymnasts, and the active flux core inside electrical solder wire.

What is Pitch?

Pitch is a thick, dark, viscous or solid residue obtained by heavily boiling down or pyrolyzing organic tars. Historically, there were two main types:

- **Pine Pitch (Wood Pitch / Brewer's Pitch):** Made by boiling down pine tar (the charred, smoky sap extracted from kiln-burning pine wood stumps). Boiling drives off the volatile oils until a thick, black, tarry waterproof glue remains.
- **Coal-Tar / Petroleum Pitch (Bitumen / Asphalt):** The heavy, black aromatic residue left after the distillation of coal tar or crude oil.
- **Properties:** Pitch is much darker, more flexible, and more tenacious than pure rosin. It softens with heat and cools to a tough, sticky, impermeable black sealant.
- **Historical Uses:** Caulking the hulls of wooden sailing ships ("paying the seams"), lining wooden beer casks (brewer's pitch), and waterproofing roofs and pipes.

Why Early Scientists Combined Them with Beeswax

In early electrochemical apparatus like Cruickshank troughs and Leyden jars, neither ingredient was used alone:

- **Pure Rosin** is too **brittle** on its own—it cracks as wood expands or metal plates flex.
- **Pure Beeswax** is too **soft** and has a low melting point (~62°C–64°C).
- **Pure Pitch** can be too **soft and tacky** at room temperature.

By melting **rosin, pitch (or beeswax), and an inert mineral filler (like brick dust, plaster of Paris, or red ochre)** together, 18th- and 19th-century builders created "**Philosophical Cement**" (or electrical cement). The rosin provided adhesion to wood and metal, the beeswax/pitch added plasticity to resist thermal shock cracking, and the powder filler gave it compressive bulk to fill gaps permanently without shrinking.

User prompt: Isn't shellac something that was available in the 19th century?

Response: **Yes, shellac was widely available and heavily used throughout the 19th century.**

It was one of the most important natural insulating and binding materials of the early electrical era, imported in massive quantities from India by the British East India Company.



What Shellac Is

Unlike pine-derived rosin or pitch, shellac is an animal byproduct. It is a natural resin secreted by the female **lac bug** (*Kerria lacca*) on trees in India and Southeast Asia. The crusty resin harvest ("sticklac") was washed, refined, and stretched into thin sheets that were broken into flakes.

How 19th-Century Scientists Used Shellac

Shellac had outstanding dielectric strength, moisture resistance, and chemical stability, making it a staple across laboratories:

- **Electrical Insulation & Wire Enamel:** Dissolved in high-proof ethanol ("spirits of wine"), liquid shellac was painted onto copper wires, electroscope rods, and glass friction machines to prevent high-voltage charge leakage.
- **Sealing Wax & Cements:** Combined with Venice turpentine and pigments (like vermilion), shellac was the primary ingredient in high-grade **sealing wax** and rigid lab cements used to seal glass tubes, Leyden jars, and galvanic apparatus.
- **Waterproofing Troughs & Woodwork:** Instrument makers painted the interior wooden frames of early batteries, galvanometer housings, and coil bobbins with shellac varnish to prevent acid moisture from soaking into the grain.

Shellac vs. Rosin/Pitch for Trough Batteries

While shellac was widely available, builders usually chose **rosin-and-pitch cements** over shellac for sealing trough partitions:

Property	Pine Rosin / Pitch Cements	Shellac
Gap-Filling Bulk	Excellent. Melts into a thick, cheap, gap-filling mastic that cools solid in minutes.	Poor. Thicker layers shrink and crack as solvent evaporates.
Cost in 1800	Extremely cheap; locally sourced from pine forestry.	Imported luxury commodity; substantially more expensive.
Brittleness	Easily plasticized with beeswax or pine pitch to resist vibration.	Hard and glass-brittle when melted without extensive plasticizers.
Acid Resistance	Impervious to dilute H ₂ SO ₄ and HNO ₃ .	Resistant to dilute acid, but can soften slowly in some alkaline environments.

19th-century builders typically used shellac as a **painted insulating varnish or fine glass-to-metal sealant**, while reserving cheap, heavy-bodied rosin/pitch mixtures for **caulking deep grooves and large wooden battery troughs**.

User prompt: What’s a recipe for 19th century battery cement?

Response: The standard 19th-century formula used by instrument makers and natural philosophers to seal trough partitions, galvanic apparatus, and Leyden jars was known as **Electrical Cement** (or *Philosophical Cement*).

Standard Electrical / Battery Cement (By Weight)

- **5 parts Pine Rosin (Colophony)** — The adhesive binder.

- **1 part Beeswax** — The plasticizer (prevents the rosin from cracking).
- **1 part Inert Powder Filler** (historically **Red Ochre**, finely ground **Plaster of Paris**, or **Brick Dust**) — Adds structural body and reduces thermal shrinkage.

Alternate Historic Variations

- **For High Flexibility / Vibration Resistance (Pitch Heavy):**
 - 4 parts Rosin
 - 1 part Pine Pitch
 - 1 part Beeswax
 - 1 to 2 parts Finely sifted brick dust or dry Spanish red oxide.
- **For Extra Adhesion to Smooth Surfaces (Glass / Smooth Metal):**
 - Replace half of the beeswax with **Canada balsam** or **Venice turpentine**.

Preparation & Application Protocol

1. **Dry the Filler:** Heat the red ochre, brick dust, or plaster of Paris in a dry pan or oven to drive off all residual moisture. *(If damp powder is added to hot molten rosin, it will froth violently and trap air pockets).*
2. **Melt the Base:** Melt the rosin and beeswax together over low heat (ideally in a double boiler or controlled heat bath around 100°C–120°C / 212°F–250°F) until completely liquefied.
3. **Incorporate the Powder:** Slowly stir in the warm, dried powder filler until the mixture is smooth and completely homogenous.
4. **Application:**
 - Warm the wooden trough grooves and metal plates slightly before applying the cement. If applied to freezing-cold metal, the mastic quenches instantly and fails to wet the surface properly.
 - Pour or trowel the hot cement into the slots and press the divider plates in firmly.
 - Smooth the joint fillets using a heated iron spatula or hot metal rod. The cement sets completely rigid as soon as it cools to room temperature.

Sources

1. [The microscopist : a manual of microscopy and compendium of the microscopic science : micro-minerology, micro-chemistry, biology - Internet Archive](#)

User prompt: Brick dust is a ridiculous price because it is used in ceremonies of some kind. Red ochre comes back as paint, or bags of iron oxide. It is iron oxide plus silica and clay. When I search

for clay powder I get bentonite, so I'm wondering if I could use crushed bentonite kitty litter. Plaster of Paris would be next cheapest. I have kitty litter

Response: **Yes, crushed bentonite kitty litter will work, but Plaster of Paris is the superior, easier, and historically authentic choice.**

Using Bentonite Kitty Litter: Pros & Pitfalls

If using plain, unscented, additive-free clay litter:

- **The Swelling Problem:** Bentonite is a smectite clay engineered to swell significantly when in contact with water. In a hydrophobic matrix of molten rosin and beeswax, the individual clay particles are largely encapsulated. However, if micro-cracks form or edge boundaries are exposed to aqueous battery electrolyte, bentonite can absorb water, swell, and compromise the seal over time.
- **Moisture Foaming:** Raw bentonite holds bound atmospheric moisture. If added directly to molten rosin (~120°C), it will cause violent boiling and foaming, trapping micro-voids in the cement. **It must be baked bone-dry** in an oven at 150°C (300°F) for an hour before mixing.
- **Grinding Effort:** Uncrushed litter grains are too coarse (aggregates will prevent the cement from packing tightly into narrow slots). It must be pulverized to a fine flour using a mortar and pestle or shop mill.

Why Plaster of Paris is the Better Choice

Plaster of Paris (calcium sulfate hemihydrate) was one of the three primary fillers cited directly in 19th-century electrical manuals (along with red ochre and Spanish whiting):

- **Ultra-Fine Texture:** Already milled to micron-scale particle sizes, allowing the molten cement to flow into tight joint channels.
- **Dimensional Stability:** Does not swell or expand in the presence of water or acidic/saline electrolytes once locked into the wax/resin binder.
- **Thermal Shock Resistance:** Adds substantial compressive strength and prevents the rosin from shrinking and pulling away from metal plates as it cools.
- **Cost:** Available at any hardware or craft store for a few dollars.

Other Cheap, Readily Available Fillers

If looking around the shop or local garden/hardware center, several everyday minerals act as inert fillers for electrical cement:

- **Chalk / Whiting / Agricultural Limestone (Calcium Carbonate):** Historically called "Spanish Whiting." Non-swelling, cheap, and very fine.
- **Play Sand / Quartz Silica Flour:** Extremely inert and impervious to all battery acids, though harder to find in fine flour grades.

- **Diatomaceous Earth:** Very light, non-reactive, and already an ultra-fine powder (bake dry before use).

Baking the filler dry before stirring it into the molten resin is the critical step to prevent bubbling and ensure a dense, leak-proof seal in the trough.

User prompt: I will use plaster of Paris . What is a Daniel cell?

Response: The **Daniell cell** is a two-fluid primary electrochemical cell invented in 1836 by British chemist John Frederic Daniell. It was the first battery to produce a steady, reliable, unvarying direct current over long periods, making it the workhorse power source of the early telegraph industry.

The Problem It Solved: Cathode Polarization

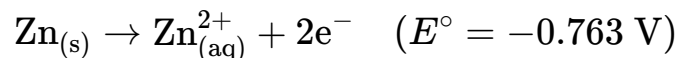
Earlier single-fluid batteries (like the Voltaic pile and Cruickshank trough) put zinc and copper into the same acid or salt solution. Under load, hydrogen ions reduced directly on the copper cathode ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$), creating an insulating blanket of gas bubbles that caused voltage to collapse within minutes.

Daniell solved this by **eliminating hydrogen gas evolution entirely**. Instead of reducing protons, the cell uses a second electrolyte containing copper ions (Cu^{2+}), so that discharge deposits solid copper onto the cathode rather than generating gas.

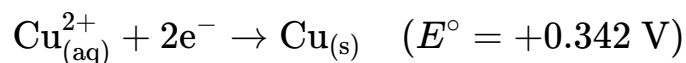
Anatomy & Cell Reactions

A Daniell cell consists of two separate half-cells separated by a porous barrier (an unglazed earthenware pot or salt bridge) to allow ion transfer while preventing rapid liquid mixing:

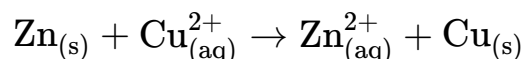
- **Anode (Negative):** A zinc plate immersed in dilute **zinc sulfate** (ZnSO_4) or dilute sulfuric acid (H_2SO_4).



- **Cathode (Positive):** A copper plate immersed in a saturated **copper(II) sulfate** (CuSO_4) solution, often kept saturated with blue bluestone crystals.



- **Overall Reaction:**



The standard open-circuit potential is remarkably stable at $\approx 1.10 \text{ V}$. Because of this steady output, the Daniell cell was historically used as the international reference standard for the definition of the **volt**.

The Two Major Historical Configurations

- **1. Porous Pot Type (Standard Lab Design):** An unglazed ceramic cup sits inside an outer glass or copper container. One half-cell is placed inside the cup (usually zinc in ZnSO_4 / acid), and the other half-cell occupies the outer vessel (copper in saturated CuSO_4).
 - **2. Gravity Cell (Crowfoot Cell):** Developed by C.F. Varley and J.B. Fuller in the 1850s, this eliminated the porous pot entirely. A dense, concentrated CuSO_4 solution sits naturally on the bottom of a glass jar around a copper "leaf" electrode, while a lighter, less dense ZnSO_4 solution floats on top around a claw-shaped zinc "crowfoot" anode. As long as the jar is undisturbed and drawn under continuous light current, the density boundary prevents copper ions from reaching the zinc.
-

User prompt: Can the separator be cast from plaster of Paris?

Response: **Yes, cast Plaster of Paris works well as a porous separator**, and it has strong historical precedent. In 1837, British physician Golding Bird developed a famous variant of the Daniell cell using a cast Plaster of Paris partition instead of an unglazed earthenware pot.

How It Functions Electrochemically

When Plaster of Paris (calcium sulfate hemihydrate) cures with water into gypsum (calcium sulfate dihydrate), it forms an interlocking, needle-like crystalline matrix ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).

- **Microporosity:** The space between crystals creates an interconnected network of microscopic pores that soak up electrolyte, allowing sulfate (SO_4^{2-}), zinc (Zn^{2+}), and sodium/hydronium ions to migrate freely.
- **Diffusion Barrier:** The tortuous pore structure slows bulk hydraulic mixing and diffusion, keeping the dark blue Cu^{2+} ions from reaching the zinc anode while the cell is active.

Practical Casting Tips for a DIY Cell

- **Target Thickness (3 mm to 6 mm):**
 - **Too thin (<2 mm):** Very fragile and prone to cracking under water pressure.
 - **Too thick (>8–10 mm):** Drastically increases internal ionic resistance (IR drop), reducing usable output current.
 - A 3–5 mm cast plate hits the sweet spot for durability versus internal resistance.
- **Water-to-Plaster Ratio:** Mix slightly wet (around a 1:1 to 1.2:1 water-to-plaster weight ratio). Using adequate water ensures open, connected porosity throughout the cured block rather than an ultra-dense, impermeable matrix.
- **Pre-soaking Before Use:** Never place a dry plaster diaphragm into active battery electrolytes. **Pre-soak the cured plaster partition in clean water or dilute salt/zinc solution for a few hours first.** If installed dry, it will greedily wick up the copper sulfate

solution by capillary action, drawing copper ions directly through the pores to the other side before the cell even starts running.

- **Mechanical Reinforcement:** Plaster is brittle in thin sheets. Casting it over a piece of open plastic mesh, coarse cheesecloth, or fiberglass drywall tape embedded in the center gives it tensile strength and prevents catastrophic cracking if handled or bumped.
- **Lifespan Limit:** Gypsum is slightly soluble in water (~2 g/L) and softens in strong acids over time. For neutral salt or dilute sulfate electrolytes, a plaster separator will easily last through weeks of experimentation, but it is not intended as a permanent multi-year installation like vitrified earthenware.

Sources

1. [History of the battery - Wikipedia](#)
-

User prompt: The copper cathode could be on the inside of the pot correct? Just wondering about using copper pipe

Response: **Yes, the copper cathode and CuSO_4 solution can be placed on the inside of the porous cup, but using a copper pipe as the *outer* vessel is actually the most authentic and efficient design.**

1. Copper on the Inside (Inverted Layout)

- **Setup:** A copper tube or rod sits inside the porous cup filled with CuSO_4 , while the outer bath contains the zinc anode and ZnSO_4 (or NaCl / dilute acid).
- **Electrochemical Function:** The cell works normally at ~1.1 V.
- **Drawback:** The interior volume of the porous pot is small. This limits the volume of CuSO_4 and surface area of the copper cathode, meaning Cu^{2+} ions deplete quickly unless you frequently add solid bluestone crystals into the pot.

2. Copper Pipe as the Outer Vessel (Original 1836 Layout)

John Frederic Daniell's original patent actually used a **copper cylinder/pipe as the container itself:**

- **Double Duty:** A capped copper pipe serves as both the **outer container** and the **active cathode**.
- **High Surface Area & Low Resistance:** The entire inner circumference of the pipe acts as the cathode, surrounding the central porous cup symmetrically. This maximizes active surface area and minimizes ionic path length (IR drop).

- **Reservoir Space:** The outer space between the pipe and the porous pot leaves plenty of room for saturated CuSO_4 solution and a perforated shelf of copper sulfate crystals near the top. As copper plates out, dissolving crystals naturally maintain saturation via density circulation.

Practical Fabrication Note with Copper Pipe

If using standard copper plumbing pipe (e.g., 1.5" or 2" Type M/L/DWV):

- Solder a standard copper end-cap to the bottom (or seal it into a 3D-printed base with epoxy/resin).
- Place the cast plaster separator (or ceramic cup) with the zinc rod in the center.
- Fill the outer annulus with CuSO_4 and connect directly to the copper pipe's outer rim for the positive terminal.

User prompt: Could I spot weld zinc foil to thin copper sheets? Or what is the best way to join the two together? I have a small spot welder for battery strips. How would it have been done in the 19th century?

Response: Spot welding zinc foil directly to copper sheet with a standard DIY pulse/capacitive battery spot welder will generally not work well, but soft soldering is straightforward, highly effective, and historically authentic.

Why Small Battery Spot Welders Struggle Here

Battery tab spot welders work by passing a brief, intense burst of current through high-resistance nickel strip to melt it at the contact interface. Joining zinc to copper creates two major physical hurdles:

- **High Thermal & Electrical Conductivity:** Copper has nearly four times the electrical conductivity and six times the thermal conductivity of nickel. It shunts the current away and conducts heat out of the contact point so fast that a low-power DIY welder cannot generate the localized resistance heat needed to fuse the metals.
- **Vaporization & Burn-Through:** Zinc melts at **419.5°C** and boils at **907°C**, while copper melts at **1085°C**. By the time enough heat is pumped into the copper to make it weldable, thin zinc foil will vaporize in a burst of white zinc oxide smoke, blasting a hole straight through the foil rather than forming a weld nugget.

(If you ever attempt spot welding with zinc, avoid inhaling the fumes, as vaporized zinc causes metal fume fever).

How It Was Done in the 19th Century

In William Cruickshank's original 1800–1802 trough batteries, the plates were joined using **soft soldering (tin-lead alloy)**:

- **Solder Seam / Lamination:** 19th-century builders tinned the copper sheet and zinc plate using a charcoal-heated copper soldering iron, killed spirits (zinc chloride flux), and common solder (50/50 Sn-Pb).
- **Face-to-Face vs. Edge Joining:**
 - In Cruickshank's design, equal-sized square plates of zinc and copper were soldered together **face-to-face** (or sweat-soldered along their perimeter edges) to form a rigid, solid bimetallic partition.
 - In Babington/Wollaston plunge batteries, the plates were joined at the top by a small soldered arch or strip of copper/brass that bent over the wooden suspension bar.

Best Methods for Your Shop

1. Sweat Soldering (Best Electrochemical & Mechanical Bond):

- **Flux:** Use an aggressive flux for zinc (rosin flux is too weak on zinc oxide). Traditional **zinc chloride flux** (historically "killed spirits") or a modern liquid water-soluble acid flux cleans both metals instantly.
- **Process:** Lightly tin the contact edge of the copper sheet with standard solder (60/40 Sn-Pb or lead-free Sn-Cu). Lay the clean zinc foil over it, apply a drop of flux, and touch the joint with a medium-temperature iron (~250°C–300°C).
- **Tip:** Do not overheat the zinc—solder flows onto zinc very quickly once the oxide is stripped by flux. Rinse with water afterward to remove flux residue.

2. Mechanical Fastening / Riveting (Fast & Zero Heat):

- Use small copper, brass, or aluminum pop rivets or eyelets to crimp the zinc foil and copper sheet together along the top dry tab.
- Applying firm mechanical clamping pressure between clean bare faces provides low electrical contact resistance without any risk of melting the foil.

3. Nickel-Strap Sandwich (Using Your Spot Welder):

- If you want to use your spot welder, you can sandwich the metals: spot weld a nickel strip to another nickel strip clamped tightly around the foil tab, or use the spot welder to tack nickel strip directly to thin nickel-plated contact shims rather than welding bare zinc to bare copper.

User prompt: The method you outlined only joins the plates along one edge?

Response: Yes—and historically, joining only along the edges was standard practice.

In early 19th-century trough batteries, the plates did **not** need full face-to-face contact to function as bipolar dividers.

How 19th-Century Builders Joined Bipolar Pairs

- **Perimeter/Edge Soldering:** Cruickshank and later instrument makers typically placed a square of zinc and a square of copper back-to-back and ran a bead of solder along the top and side edges (or just the top and bottom corners).
- **Why Full Face Lamination Wasn't Needed:**
 - **Low Current Densities:** These cells produced small currents (milliamps to a few hundred milliamps). Because both copper and zinc are excellent electrical conductors, electrons flow easily from the entire zinc face, through the sheet, to the soldered edge seam, and back across the copper plate with negligible voltage drop.
 - **Mechanical Flatness:** Early rolled or cast metal sheets were rarely optically flat. Soldering only the edges was much faster, consumed far less tin/lead solder, and prevented trapped flux pockets between the plates.

Modern DIY: Top-Edge vs. Full-Face Lamination

For a DIY cell, you have three options depending on your design:

Method	How It's Done	Pros / Cons
Top-Edge Seam / Strap	Solder or clamp the zinc and copper together only along the top tab/rim above the waterline.	Easiest and cleanest. Keeps the bimetallic joint completely dry, preventing local galvanic corrosion at the solder seam.
Edge-Perimeter Soldering (Historical Cruickshank)	Run a thin bead of solder along 2–3 outer edges of the paired plates.	Forms a single rigid plate ready to slide into trough grooves and seal with pitch/cement.
Full-Face Sweat Soldering	Tin the entire back of both sheets, clamp them in a vise or flat plates, and heat with a torch until solder flows across the whole area.	Maximum rigidity and lowest internal resistance, but requires significant heat and can easily warp thin foils or melt zinc.

If you use solid 3D-printed partition walls with separate slots for the zinc and copper, **joining them along the top dry edge via a soldered tab or bent nickel strap is all you need**—the electrical conduction across the top bridge will carry the full current without measurable loss.

User prompt: So clean with flux, melt a bead of solder onto one edge of the copper (inside the edge?) , then press the zinc on and slowly run a soldering iron across the edge to be joined to melt the solder

Response: **Yes, that is the exact essence of sweat soldering, but with zinc, heat management and iron placement require one specific technique.**

Because zinc foil melts at a relatively low temperature (419.5°C) and conducts heat quickly, pressing a hot iron directly onto the thin zinc face can melt or tear the foil before the solder underneath flows.

Step-by-Step Technique for Zinc-to-Copper Sweat Soldering

1. Surface Prep & Tinning the Copper:

- Clean both metal surfaces with fine abrasive (Scotch-Brite or steel wool) until shiny.
- Apply a drop of liquid acid/zinc chloride flux along a 3–5 mm strip near the top edge of the copper.
- Run your iron across that strip with a small amount of solder to create a thin, uniform, pre-tinned silver band on the copper (keep it flat, avoiding tall lumps).

2. Flux the Zinc:

- Clean the mating edge of the zinc foil.
- Apply a thin film of flux to the back edge of the zinc (where it will touch the tinned copper band).

3. Align & Clamp:

- Overlap the zinc edge directly onto the tinned copper band.
- Hold the pieces firmly in place with a wooden stick, dry scrap wood, or metal tweezers so they cannot shift during heating.

4. Heat Through the Copper (The Pro Trick):

- **Apply the soldering iron tip to the thicker copper side** (or just adjacent to the zinc edge on the copper sheet) rather than directly burning through the zinc foil.
- Copper will absorb the heat and conduct it through the joint, melting the pre-tinned solder cushion.
- As soon as you see the solder flash/liquefy and wet the seam under the zinc, glide the iron along the seam and remove the heat immediately.

5. Cool & Rinse:

- Keep light pressure on the zinc for 3–5 seconds until the solder solidifies from shiny liquid to a dull solid.
- **Rinse thoroughly with warm water** to wash away all acid flux residue, preventing white crusty corrosion at the joint later.

User prompt: I suppose the other edges would be held close by the slot. If I do this I would be aiming for a historical battery. For funsies

Response: **Exactly—that is the beauty of the Cruickshank design.**

Once the paired pair of plates slides into the side and bottom dados of the wooden trough, the surrounding electrical cement seals and holds the remaining three edges firmly in place.

How the Assembly Locks Together

1. **Top Solder Seam:** Joining the zinc and copper along the top rim keeps them electrically linked, prevents electrolyte from wicking down between the faces, and makes handling the pair as a single rigid unit easy.
2. **Groove Retention:** The side and bottom edges slide directly into the kerfs cut across the floor and walls of the wooden box.
3. **Cement Potting:** Molten electrical cement (rosin + beeswax + dried Plaster of Paris) is poured directly into the groove joints. It flows around both metal edges, potting and sealing the bottom and sides into a continuous, liquid-tight barrier between adjacent cells.

Key Historical Details for Authenticity

- **Wood Choice:** Historically, well-seasoned mahogany, oak, or walnut was preferred because it resists warping when moist. The entire interior of the wood box was coated in hot resin/varnish to seal the grain before installing plates.
- **Plate Orientation:** Every pair faces the exact same direction (e.g., Copper → Zinc | Electrolyte | Copper → Zinc). This puts all the cells in series, adding ~1.0 V to 1.1 V per compartment.
- **Terminal Leads:** The end plates get heavy copper wire terminals soldered directly to the exposed end faces, often routed to small brass binding posts or simple wooden side notches.

User prompt: Current delivery would be limited by the area of the plates?

Response: **Yes, the maximum current a trough battery can deliver is directly proportional to the submerged surface area of the plates.**

This relationship between plate size, cell count, and electrical output was one of the first major discoveries of early 19th-century electrochemistry.

Why Surface Area Limits Current

The peak and sustained current delivery are dictated by electrochemical and transport limits that scale directly with active area (A):

- **Ohmic Internal Resistance (R_{int}):** The bulk electrolyte resistance of each cell is inversely proportional to plate area:

$$R_{\text{cell}} = \frac{\rho \cdot d}{A}$$

Doubling the submerged area halves the cell's internal resistance, which cuts the internal voltage drop (IR) under load in half.

- **Limiting Current Density (i_{limit}):** Electrochemical reactions have an inherent rate limit per unit area ($i = I/A$ in mA/cm^2):
 - **In Neutral Brine (NaCl):** The cathode is throttled by dissolved oxygen diffusion ($\sim 0.1\text{--}1.0 \text{ mA}/\text{cm}^2$). To draw 100 mA without crashing cell voltage into hydrogen evolution, you physically need $\sim 100\text{--}1000 \text{ cm}^2$ of active copper area.
 - **In Dilute Acid (H_2SO_4):** Hydrogen evolution allows much higher rates ($\sim 10\text{--}50 \text{ mA}/\text{cm}^2$), but high current densities accelerate zinc surface passivation and create severe local gas blinding. A larger plate area keeps the *local* current density low, delaying polarization.

The Historical "Davy vs. Children" Rules

Early researchers like Humphry Davy and John George Children realized that battery dimensions dictated what experiments they could run:

Dimension Changed	What Increases	Historical Demonstration
Increase Number of Cells (N) (Many small plates)	Voltage ($V_{\text{total}} = N \cdot V_{\text{cell}}$)	Decomposing chemical compounds, making sparks jump across gaps, giving strong physiological shocks.
Increase Plate Area (A) (Few massive plates)	Current ($I_{\text{max}} \propto A$)	Heating and melting thick platinum or iron wires, igniting gunpowder, magnetic deflection.

Cruikshank troughs built for general laboratory work typically used plates around **3" × 3" (7.5 cm × 7.5 cm)** or **4" × 4" (10 cm × 10 cm)**, which provided enough surface area to sustain 50–200 mA loads for basic electrolysis and magnetic experiments without collapsing under rapid polarization.

User prompt: Wait is the active area of copper the area of a single plate or of the entire battery?

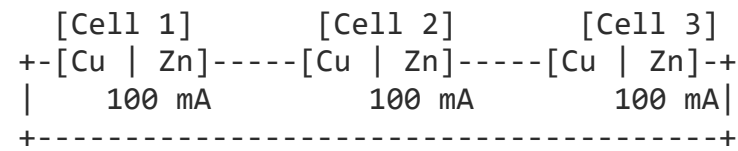
Response: **It is the area of a single plate (or single cell), NOT the sum of the entire battery.**

Because a trough battery connects all its cells in **series**, every individual cell carries the exact same total electrical current.

The Series Circuit Bottleneck

In a series string:

- The total current flowing through the circuit must pass through every cell one after another.
- If your trough has 10 cells, and you draw 100 mA from the battery terminals, that exact same 100 mA is flowing across Cell 1, Cell 2, Cell 3... all the way to Cell 10.
- Therefore, the **current density** ($i = \frac{I}{A}$) at the copper surface in *each* cell is determined solely by the submerged surface area of that **single plate**.



If the single plate area is 25 cm², drawing 100 mA forces a current density of 4 mA/cm² on every single plate. Adding more cells in series increases the **total voltage**, but it does nothing to alleviate the current density on any individual plate.

Series vs. Parallel Summary

Battery Configuration	Total Voltage	Max Current Capability	Determining Factor
Series (Standard Trough)	$V_{\text{total}} = N \times V_{\text{cell}}$	Unchanged (Same as 1 cell)	Area of a single plate
Parallel (Plates wired side-by-side)	$V_{\text{total}} = V_{\text{cell}}$ (~1.0 V)	$I_{\text{total}} = N \times I_{\text{cell}}$	Combined total area of all plates

If you want the battery to supply a higher sustained current without rapidly polarising, you must physically increase the **width and height of the individual plates**, rather than just adding more compartments to the trough.

User prompt: So a 4" x 4" sheet loses some area at the top to stay dry and around the edges. So let's say 80 cm^2. Using your numbers above I get just over 80 amps in acid and 88 mA in brine. That seems wildly different

Response: Taking the current density ranges mentioned earlier:

For **brine (0.1 to 1.0 mA/cm²)**:

At 1.0 mA/cm² × 80 cm² = 80 mA

$$\text{At } 1.1 \text{ mA/cm}^2 \times 80 \text{ cm}^2 = 88 \text{ mA}$$

For **acid (10 to 50 mA/cm²)**:

$$\text{At } 10 \text{ mA/cm}^2 \times 80 \text{ cm}^2 = 800 \text{ mA (0.8 A)}$$

$$\text{At } 50 \text{ mA/cm}^2 \times 80 \text{ cm}^2 = 4000 \text{ mA (4.0 A)}$$

An arithmetic unit conversion slip caused the acid estimate to appear as 80 A instead of 0.8 A to 4 A.

Even with the corrected units, the difference between **~80 mA** (brine) and **~800 to 4,000 mA** (acid)—a factor of 10× to 50×—is genuine, and stems directly from the reaction mechanisms:

- **In Brine (Mass-Transport Limited):** The cathode depends on ambient dissolved oxygen (O₂), which is present only in trace amounts (around 8 mg/L or ~ 0.25 mM). The reaction rate is strictly bottlenecked by how quickly oxygen molecules physically diffuse through stagnant water to hit the copper surface.
- **In Acid (Reactant-Abundant):** The cathode reduces hydronium ions (H⁺), which are present at enormous concentrations (1 M to 2 M, or nearly 10,000 times more concentrated than dissolved oxygen). Because reactant ions are packed right against the metal surface, current can flow orders of magnitude faster before becoming diffusion-limited.

User prompt: What are some historical dimensions of Cruikshank cells ?

Response: Historical Cruickshank trough batteries from the early 1800s were built in a few standardized configurations depending on whether they were intended for high-tension sparks (many small plates) or high-current chemical deflagration (fewer large plates).

Standard Laboratory Sizes (1800–1810)

- **The Small / High-Tension Model (Davy's Multi-Cell Troughs):**
 - **Plate Size:** 1.5" × 1.5" to 2" × 2" (approx. 3.8 cm to 5.0 cm square).
 - **Cell Count:** Typically **50 to 100 pairs** in a single trough (often 24" to 36" long).
 - **Inter-Plate Gap (Compartment Width):** 0.25" to 0.375" (~6 mm to 9 mm).
 - **Purpose:** Built to generate hundreds of volts in series for physiological experiments, electric arcs, and decomposing salts.
- **The Standard Medium Laboratory Workhorse:**
 - **Plate Size:** 3" × 3" or 4" × 4" (approx. 7.5 cm to 10 cm square).
 - **Cell Count:** Usually **20 to 25 pairs** per wooden box.

- **Inter-Plate Gap: 0.375" to 0.5"** (~9 mm to 13 mm).
- **Trough Dimensions:** Roughly 24" to 30" long, 5" wide, and 4" to 5" deep (internal).
- **Purpose:** The everyday bench standard for electrolysis of water and metal electrodeposition.
- **The "Deflagrator" / High-Current Sizes:**
 - **Plate Size: 6" × 6" to 8" × 8"** (approx. 15 cm to 20 cm square).
 - **Cell Count: 10 to 20 pairs** per trough.
 - **Inter-Plate Gap: 0.5" to 0.75"** (~13 mm to 19 mm) to allow sufficient acid volume and gas venting.
 - **Purpose:** Igniting gunpowders, melting thick metal wires, and producing continuous arc lighting.

Structural Construction Specs

- **Wood Box Thickness:** Typically **0.75" to 1.0"** (19 mm to 25 mm) thick seasoned mahogany, walnut, or oak to prevent warping from acid moisture.
 - **Kerf / Groove Depth:** Side and bottom dados were cut **0.125" to 0.187"** (3 mm to 5 mm) deep into the wood walls to seat the bimetallic plates firmly before pouring the resin cement.
 - **Liquid Level:** Boxes were filled to within **0.5" to 0.75"** (12 mm to 19 mm) of the top rim, leaving the soldered top edge of the plates dry.
-

User prompt: What thickness of metal would be used?

Response: In 19th-century Cruickshank troughs, the zinc and copper plates were intentionally made with **unequal thicknesses** because they served very different roles:

- **Zinc Plate (The Sacrificial Anode): 1.5 mm to 3.0 mm** (approx. 1/16" to 1/8" / 11 to 16 gauge).
 - Because zinc is chemically consumed during discharge and constantly dissolves away into the acid, thin zinc foils would perforate and fail after just a few runs.
 - A thick zinc plate provided structural rigidity across the wooden groove and enough sacrificial metal mass to survive multiple cleanings and acid refills before needing replacement.
- **Copper Plate (The Insoluble Cathode): 0.5 mm to 1.0 mm** (approx. 1/32" to 1/16" / 18 to 24 gauge, historically called "stout sheet copper" or 16–24 oz roofing/brazier copper).
 - Copper does not dissolve during normal discharge; it merely serves as an electron conductor and gas evolution substrate.

- It only needed to be thick enough to resist crumpling, stay flat against the zinc, and provide structural stiffness for the slot seal.

Combined Bipolar Pair Thickness

When sweat-soldered back-to-back, the total paired sandwich was typically **2.0 mm to 4.0 mm** thick (~3/32" to 5/32").

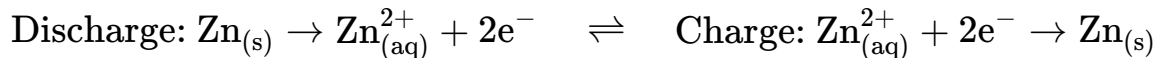
This combined thickness gave the divider enough stiffness to span a 3" to 5" wide trough box without bowing, allowing standard wood saws of the era to cut a matching ~1/8" (3 mm) kerf into the trough walls to receive the plate assembly.

Sources

1. "Historic Papers in Electrochemistry"

User prompt: Hey what changes to make a zinc anode part of a rechargeable cell? I have this niggling memory that we discussed zinc anode batteries before with graphite felt cathodes

Response: To transition zinc from a primary, single-use anode into a truly reversible, rechargeable electrode, the core electrochemical principle shifts from **sacrificial dissolution** to a clean cycle of **dissolution on discharge and uniform electrodeposition (plating) on charge**:



Making this reversible in practice requires addressing several fundamental failure modes of zinc:

1. The Three Main Hurdles of Rechargeable Zinc

- **Dendrite Growth:** During recharging, zinc ions do not naturally plate back as a smooth, flat sheet. Instead, electric fields concentrate at microscopic high points, causing tree-like metallic needles (dendrites) to grow across the separator until they short-circuit the cell.
- **Shape Change & Mass Redistribution:** Zinc dissolves from high-current-density areas during discharge and replates unevenly (often pooling near the bottom edges of the plate), causing the electrode to distort and lose active area over multiple cycles.
- **Hydrogen Evolution During Charge:** Zinc's standard reduction potential (-0.76 V) sits well outside the thermodynamic stability window of water. During recharge, water reduction ($2\text{H}_2\text{O} + 2e^{-} \rightarrow \text{H}_2 \uparrow + 2\text{OH}^{-}$) competes with zinc plating, causing parasitic gassing, loss of Coulombic efficiency, and local pH shifts.

2. Practical Strategies for Reversible Zinc

- **Mild Aqueous Electrolytes (Sulfate/Acetate vs. Alkaline):**
 - While commercial alkaline cells (using concentrated KOH) suffer from massive zinc redistribution and dendrites due to highly soluble zincate $[\text{Zn}(\text{OH})_4]^{2-}$, **mildly acidic**

or near-neutral electrolytes (such as 1–2 M ZnSO_4 or $\text{Zn}(\text{CH}_3\text{COO})_2$) drastically improve cycle life.

- In mild sulfate systems (pH ~4 to 5), zinc plates out far more smoothly and forms a self-regulating interface.
- **High-Surface-Area Substrates (3D Current Collectors):**
 - Instead of plating onto a flat zinc foil, using a 3D porous conductive host (such as carbon paper, graphite felt, copper mesh, or porous brass) dramatically lowers the **local current density** (mA/cm^2).
 - Lower local current densities eliminate electric field hot-spots and promote compact, planar nucleation rather than dendrite spikes.
- **Electrolyte Additives:**
 - Small amounts of leveling agents or surface-active additives (e.g., trace polyethylene glycol/PEG, trace Mn^{2+} , or sodium dodecyl sulfate) adsorb onto fast-growing dendrite tips, forcing zinc to deposit uniformly into the valleys.

3. Pairing with Graphite Felt (The Zinc Deposition Chemistry)

When paired with a **graphite felt cathode** in an aqueous sulfate solution:

- **The Cathode Reaction:** Often operates as a deposition/redox mechanism—for instance, in a **zinc manganese deposition battery**, manganese ions in solution deposit onto the high-surface-area graphite felt as MnO_2 on charge ($\text{Mn}^{2+} + 2\text{H}_2\text{O} \rightarrow \text{MnO}_2 + 4\text{H}^+ + 2\text{e}^-$) and dissolve back into the electrolyte on discharge.
- **The Zinc Role:** The zinc plate (or a 3D zinc-plated substrate) supplies Zn^{2+} ions into the bulk sulfate solution on discharge and plates metallic zinc back out during the constant-current charge phase.

User prompt: What other chemistries were tested in trough battery in the 19th century? Lead acid cells?

Response: The "trough" form factor evolved significantly throughout the 19th century as chemists tried to overcome the rapid polarization and acid-corrosion problems of early cells. Multiple primary and secondary chemistries were deployed in trough formats:

1. The Lead-Acid Trough (Secondary / Rechargeable)

Yes, lead-acid was built into troughs.

- **The Evolution from Planté to Faure (1860s–1880s):** Gaston Planté originally made rechargeable cells by rolling lead sheets into spirals separated by rubber strips in glass jars. However, in the 1880s, when **Camille Faure** and the **Electric Power Storage (E.P.S.) Company** scaled up commercial storage batteries using pasted grid plates, they moved to

multi-plate **trough-style enclosures** (often lead-lined wooden boxes or multi-compartment vulcanized hard rubber troughs).

- **Why the Trough Suited Lead-Acid:** The rigid rectangular trough geometry allowed heavy, flat lead grids (Pb anode, PbO₂ cathode in sulfuric acid) to be stacked compactly with bottom sediment clearances ("mud ribs") to catch shedding oxide paste.

2. Multi-Cell Daniell Troughs (Telegraph Workhorses)

While Daniell cells were often individual ceramic pots, the British telegraph service adapted them directly into standardized wooden troughs to power long lines:

- **The Muirhead Trough Battery (1857):** A wooden trough divided into 5 or 10 compartments by solid slate, porcelain, or gutta-percha partitions. Each compartment held a porous earthenware pot separating copper in CuSO₄ from zinc in dilute sulfuric acid or water.
- **The Fuller Trough (Double-Fluid Trough):** Similar multi-compartment trough using porous flat tiles cemented into the frame rather than separate round pots.

3. The Poggendorff / Bichromate "Plunge" Trough (Zinc–Carbon)

Invented in the 1840s–1850s, this became the favorite high-current lab battery of the late 19th century:

- **Chemistry:** Zinc anode paired with hard carbon (retort carbon or graphite) in an aggressive electrolyte of **potassium bichromate** (K₂Cr₂O₇), **sulfuric acid**, and **water** (known as *electropon fluid*).
- **Trough Format:** Built as a **"plunge battery"** where 6 to 12 pairs of zinc and carbon plates were ganged to a single wooden beam above a multi-compartment porcelain or lead-lined trough. Lowering the rack into the trough delivered massive, non-polarizing current at **~2.0 V per cell**; cranking the rack out of the trough stopped acid corrosion instantly when experiments finished.

4. The Smee Trough (Zinc–Platinized Silver)

Invented in 1840 by Alfred Smee, this retained the simple single-fluid trough format but fixed hydrogen polarization without needing toxic depolarizing liquids:

- **Chemistry:** Zinc anode paired with a thin **silver sheet coated in finely divided platinum black** in dilute sulfuric acid (1.0 V).
- **Trough Design:** Fitted directly into standard wooden troughs. The rough platinum black dramatically lowered hydrogen overpotential, causing bubbles to detach continuously rather than forming an insulating film. It was the premier power source for Victorian electrotyping and silver electroplating.

5. Grove and Bunsen Two-Fluid Cells

- **Grove Cell (1839):** Zinc in dilute acid paired with **platinum foil in concentrated nitric acid** (HNO_3) (~1.9 V).
- **Bunsen Cell (1841):** Robert Bunsen replaced Grove's expensive platinum with cheap **compressed carbon blocks**.
- **Trough Adaptations:** While messy due to toxic nitrogen dioxide (NO_2) gas, series banks were frequently arranged in long segmented troughs for arc-lighting and early electro-smelting demonstrations.

Sources

1. [History of the battery - Wikipedia](#)
 2. [Daniell battery hi-res stock photography and images - Alamy](#)
-

User prompt: The porous pots I can easily find but they have drainage holes in them. What would be a historically appropriate way to close those holes? I assume it doesn't have to be porous.

Response: **You are completely correct—the plug does not need to be porous.** The separator's job is simply to provide enough porous surface area for ion transport through its sidewalls; the floor can be completely solid and impermeable.

Historically, 19th-century laboratory experimenters frequently had to adapt vessels, repair cracked ceramics, or seal apparatus using the materials on hand.

Historically Authentic Sealing Methods

1. Rosin & Beeswax Battery Cement (The Best Historic Choice):

- Melt standard electrical cement (**5 parts rosin, 1 part beeswax, 1 part dried Plaster of Paris**).
- Warm the clay pot base slightly so the cement doesn't freeze on contact.
- Place a small backing disc (like a piece of thick card, parchment, or wood) on the outside of the hole, then pour molten cement into the inside bottom of the pot to a depth of **3–5 mm**.
- The hot resin wicks slightly into the open pores of the warm terracotta and sets into a rigid, acid-proof, waterproof floor within minutes.

2. Gutta-Percha Plug (Mid-to-Late 19th Century Standard):

- Natural latex from *Palaquium gutta* trees was the miracle thermoplastic of Victorian telegraphy.
- Heated in hot water until soft and pliable like clay, a wad of gutta-percha was jammed into the drain hole and pressed flat on both sides. Once cold, it hardened into an impermeable, highly acid-resistant rubber plug.

3. Natural Cork Sealed with Sealing Wax:

- Taper and press a standard natural wine cork tightly into the hole.
- Shave it flush with the pot interior, then paint or pool molten shellac sealing wax or pine pitch over the cork and surrounding clay to seal the porous cork grain.

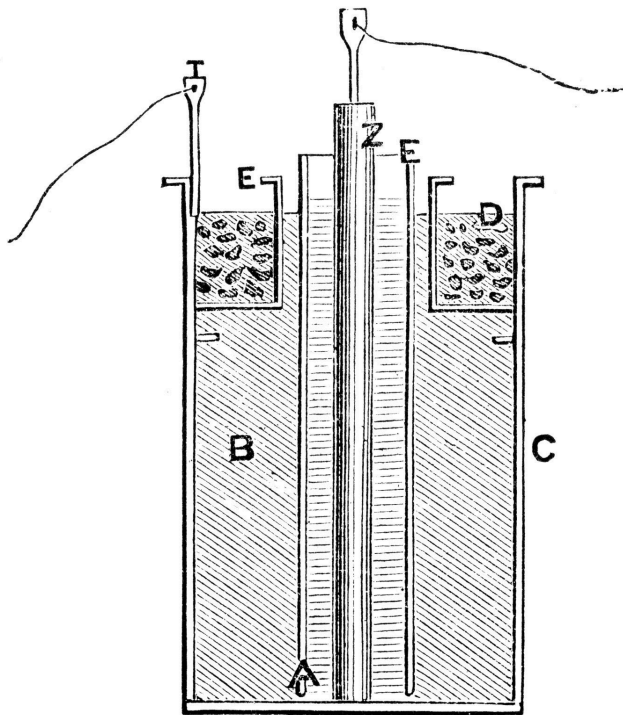
Modern Workshop Equivalents

If you want a fast, zero-fuss seal in the shop that behaves identically:

- **Plaster of Paris Plug:** Mix a stiff dab of plaster, press it into the hole, and let it cure. While the plaster itself remains slightly porous, it stops bulk liquid flow immediately.
 - **100% Silicone Caulk or Rubber Grommet/Stopper:** A dab of pure neutral/acid-cure silicone pressed into the hole creates an inert seal that resists copper sulfate and dilute acid indefinitely.
 - **Hot Melt Glue:** A thick drop of standard EVA hot glue over the hole provides a quick resinous seal that mimics pitch or gutta-percha.
-

User prompt: What was the geometry of the electrodes and the outer jar?

Response: The classic 19th-century Daniell cell used a **coaxial cylindrical geometry** (concentric cylinders nested inside one another). This radial symmetry kept the ionic path length identical in every direction, minimizing internal resistance (IR drop) and ensuring uniform current density.



Daniell's Battery.—Section of Cell.

c, Outer copper cell. B, Solution of sulphate of copper. D, Shelf for sulphate of copper. E, Porous cell. A, Sulphuric acid. z, Zinc.

The Two Historic Layouts

- **Layout A: The Original 1836 Metal-Can Design (Shown Above)**
 - **Outer Vessel (Cathode):** A tall, narrow **copper cylinder/can** (typically 3.5" to 4" diameter × 6" to 8" tall). The vessel itself served as the copper cathode.
 - **Crystal Shelf:** A perforated copper annular ring or basket suspended just below the top rim inside the outer chamber. Copper sulfate (CuSO_4) crystals sat on this shelf to maintain solution saturation as copper plated out.
 - **Porous Separator:** A tall, unglazed terracotta/earthenware cylinder (1.5" to 2" diameter) placed concentrically in the middle.
 - **Center Core (Anode):** A solid **zinc rod or heavy-walled cast zinc cylinder** (0.5" to 1" diameter) suspended down the center axis inside the porous pot, immersed in dilute acid or ZnSO_4 .
- **Layout B: The Glass Jar / Telegraph Design (Later 19th Century)**
 - **Outer Vessel:** A heavy, wide-mouth **glass or stoneware jar** (often octagonal or round).
 - **Outer Electrode (Cathode):** A thin **sheet of copper rolled into an open cylinder** lining the inside glass wall.

- **Center:** The unglazed porous pot was placed in the middle.
- **Inner Electrode (Anode):** A cast zinc rod centered inside the pot.

Why Coaxial Cylinders Were Used

- **Equalized Current Density:** A flat plate in a round jar causes current to crowd onto the nearest edges, causing uneven zinc dissolution. Concentric cylinders ensure that every square centimeter of the zinc anode faces the copper cathode at the exact same distance.
 - **Compact Footprint:** Nesting the components vertically delivered the maximum active surface area within the smallest bench footprint.
 - **Volume Distribution:** The annular (outer) ring provided a large fluid reservoir for CuSO_4 crystals and solution, while the smaller inner core kept the volume of required acid/zinc salt minimal.
-

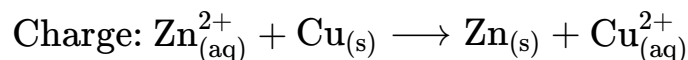
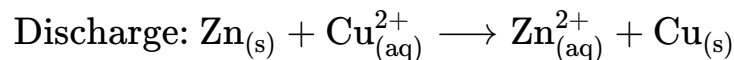
User prompt: Was the Daniel's cell rechargeable? I forget.

Response: **In theory and chemical principle, yes—but in 19th-century practical use, no.**

Historically, the Daniell cell was treated strictly as a **primary (single-use / consumable) battery**, renewed mechanically by replacing the zinc rod and adding more blue vitriol (CuSO_4) crystals rather than recharging it electrically.

The Theoretical Reversibility

Thermodynamically, the net Daniell cell reaction is fully reversible:

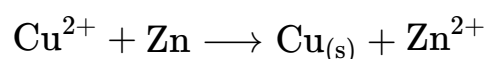


If you apply an external DC power source with a voltage greater than $\approx 1.10 \text{ V}$, copper will dissolve back into solution at the cathode and metallic zinc will plate back onto the anode.

Why It Failed as a Practical Rechargeable Cell

Victorian experimenters attempting to electrically recharge Daniell cells ran into severe physical failure modes:

- **Copper Ion Crossover & Self-Discharge:** A porous earthenware pot or plaster separator slows bulk liquid mixing, but it cannot stop continuous ionic diffusion. Over hours or days, blue Cu^{2+} ions slowly diffuse across the porous wall into the zinc chamber. The moment a Cu^{2+} ion touches the zinc anode, a spontaneous displacement reaction occurs:

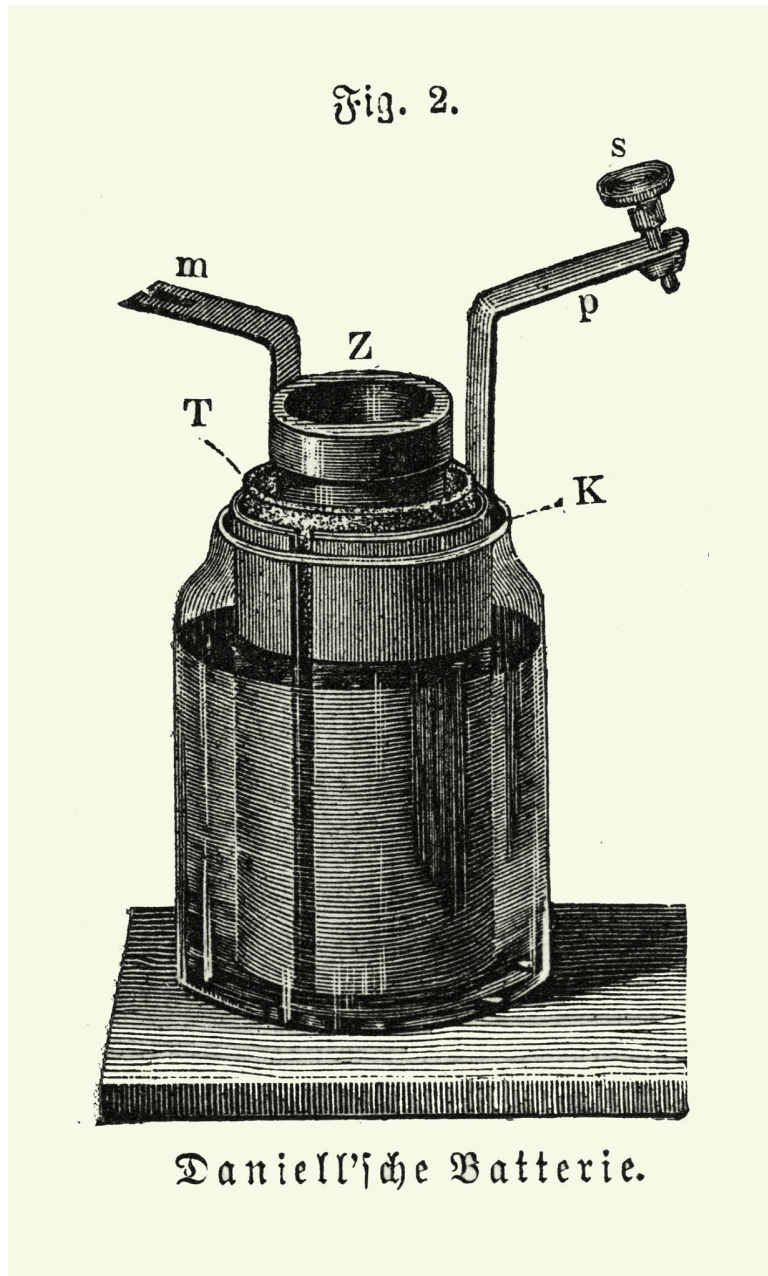


This plates a spongy, black layer of metallic copper directly onto the zinc rod, creating hundreds of short-circuited local micro-galvanic cells that rapidly corrode the zinc and kill the cell via self-discharge even when idle.

- **Zinc Dendrites During Recharge:** When forcing zinc to electroplate back onto the anode during electrical charging, it deposits as fragile, mossy needles (dendrites) rather than a flat, solid sheet. These dendritic spikes quickly grow across the separator pores, causing hard internal short circuits.
 - **Lack of DC Generators in 1836:** When John Frederic Daniell invented the cell in 1836, practical dynamos and electrical grids did not exist. There was no cheap, abundant source of DC electricity to recharge secondary batteries, making mechanical replenishment (scraping the zinc and dropping in fresh chemicals) the only viable operational workflow until Gaston Planté introduced the lead-acid accumulator in 1859.
-

User prompt: How did 19th-century telegraph stations maintain, clean, and renew Daniell cells on a routine schedule?

Response: Telegraph battery rooms housed hundreds to thousands of Daniell cells connected in series strings, demanding a rigorous, quasi-industrial maintenance regimen by dedicated battery-men ("battery attendants").



Daily & Weekly Routine Maintenance

- **Hydrometer Checks & Specific Gravity Management:**
 - In two-fluid systems, the lighter zinc solution at the top would become overly dense as zinc dissolved (ZnSO_4 concentration rising).
 - Battery attendants checked specific gravity with a Baumé hydrometer. When the top solution reached $\sim 20^\circ\text{--}25^\circ$ Baumé, they used a **battery syringe** (a large rubber bulb with a long glass tube) to siphon off the dense top liquid and replace it with fresh soft water.
- **Topping Up Blue Vitriol (CuSO_4):**

- Attendants regularly dropped fresh crystals of copper sulfate down chutes or directly onto the crystal shelf.
- The rule of thumb was to keep the crystal bed submerged to ensure the bottom cathode electrolyte remained fully saturated.

- **Creeping Salt Mitigation:**

- Sulfate salts constantly wicked up the porous pot walls and jar rims via capillary action, forming white crusty blooms that created surface leakage paths to ground.
- Rims were periodically swabbed with warm water, and the top 1–2 inches of jars were painted with **paraffin wax, mineral oil, or lard** to stop the salt creep.

Monthly / Quarterly Deep Cleaning ("The Battery Overhaul")

Every 1 to 3 months, entire battery banks were taken offline in rotation for complete mechanical teardown:

1. Scraping the Zinc Anodes:

- Attendants pulled the zinc rods/crowfeet, which were heavily caked in black, spongy copper mud (from unavoidable copper-ion crossover) and zinc oxide sludge.
- Zincs were wire-brushed, scraped down to bare metal with draw-knives, and weighed. Severely thinned zincs were melted down in a shop crucible and recast into new rods.

2. Re-Amalgamating with Mercury:

- After scraping, zincs were dipped in dilute sulfuric acid and rubbed with liquid mercury using a stiff rag until a bright, reflective liquid-metal mirror formed. This suppressed local iron-impurity galvanic action ("local action").

3. Porous Pot Leaching:

- Terracotta pots were soaked in large tubs of boiling water and weak acid for days to dissolve crystallized copper and sulfate plugs out of the ceramic pores, restoring ion permeability.

4. Copper Reclamation:

- The solid metallic copper that had plated out of solution onto the cathodes was peeled off in sheets and sold back to scrap dealers or reused in foundry melts.

Trough vs. Gravity Cell Operational Differences

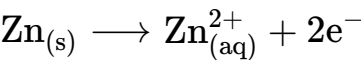
Factor	Porous Pot Trough (Muirhead / Standard Daniell)	Gravity "Crowfoot" Cell (Late 19th Century)
Separation Method	Rigid unglazed clay pot or cast divider	Fluid density alone (dense blue CuSO ₄ on bottom, clear light ZnSO ₄ on top)
Maintenance Burden	High; pots clogged with crystals and cracked if left dry.	Lower; zero pots to scrub or break.
Requirement	Could tolerate slight bench vibrations.	Had to remain completely undisturbed on shock-isolated wooden racks; vibration mixed the density boundary.
Closed-Circuit Need	Could sit on open circuit without rapid mixing.	Required continuous light current draw (closed loop) to keep the blue copper line driven down away from the zinc crowfoot.

User prompt: Ah, the shelf increased the capacity of the cell by restocking the copper sulfate. Was the sulfuric acid next to the zinc needing replenishment?

Response: No, the acid didn't run out—in fact, the opposite happened: the zinc chamber became choked with excess zinc salt and needed to be diluted.

What Actually Happened at the Zinc Anode

When the cell discharged, zinc oxidized into solution:



- **If started with dilute sulfuric acid (H₂SO₄):** The acid naturally converted into zinc sulfate (ZnSO₄). As discharge continued, the liquid around the zinc grew progressively denser and more concentrated with Zn²⁺ and SO₄²⁻ ions.
- **The Problem of Oversaturation:** Once the zinc solution became near-saturated, several issues arose:
 1. **Voltage Sag:** As [Zn²⁺] concentration spikes, the Nernst equation shows cell voltage drops (from ≈ 1.14 V down toward ≈ 1.05–1.08 V).
 2. **Internal Resistance Spike:** Extremely concentrated zinc sulfate increases the internal resistance of the liquid.
 3. **Crystallization:** White ZnSO₄ crystals would begin crusting over the zinc anode and clogging the microscopic pores of the earthenware pot.

How Attendants Handled It: Dilution, Not Addition

Instead of adding acid or salt, the standard operational protocol was to **siphon out concentrated zinc liquid and add plain water**:

1. The attendant used a rubber battery syringe to draw off the dense, heavy ZnSO_4 liquid sitting around the zinc.
2. They refilled the compartment with fresh soft water.
3. The remaining dissolved salts and migrating sulfate ions were more than enough to maintain electrical conductivity without adding any new acid.

The Telegraph Transition: Abandoning Acid Entirely

In early laboratory Daniell cells, dilute sulfuric acid was initially used around the zinc to lower internal resistance. However, acid caused **local parasitic corrosion** (the acid slowly ate the zinc metal even when the telegraph line was idle).

By the 1850s, telegraph networks largely abandoned acid in the porous pot:

- They filled the zinc chamber with **plain water** (or a very weak, dilute pinch of ZnSO_4 or MgSO_4 just to start initial conduction).
- As soon as current was drawn, the zinc dissolved, generating its own ZnSO_4 electrolyte in situ.
- This eliminated parasitic acid bubbling, extended zinc life dramatically, and simplified maintenance to two simple rules: **keep the copper side supplied with blue crystals, and periodically dilute the zinc side with water**.

Sources

1. [Daniell](#)
2. [Daniell Cell | Youtblob](#)

User prompt: I'm not sure I understand how the Daniel cells were integrated into the wooden troughs

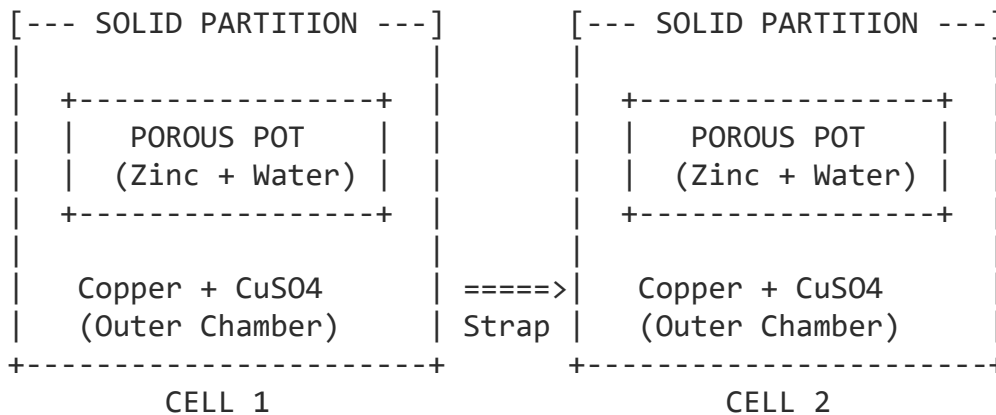
Response: In a **Cruickshank trough**, the bimetallic plates *are* the physical dividers. But a **Daniell cell** requires two different electrolytes (CuSO_4 and ZnSO_4 /water) separated by a porous barrier.

To make this work in a long wooden trough, 19th-century telegraph engineers abandoned the bipolar metal divider and used **solid, liquid-tight insulating partitions**, organizing the trough in one of two ways:

1. The Muirhead Trough Design (Most Common)

This was the standard British Post Office and telegraph trough:

- **Solid Cell Compartments:** A long wooden box (usually teak or mahogany lined with pitch) was divided into 5 or 10 completely watertight compartments using solid partitions of **slate, porcelain, or gutta-percha**.
- **Porous Pot Inside Each Cell:** Inside each watertight compartment sat a flat, rectangular unglazed earthenware porous pot.
- **Liquid Separation:**
 - **Outer chamber** (around the pot): Filled with saturated copper sulfate (CuSO_4) solution, blue vitriol crystals, and a flat sheet of copper.
 - **Inner pot:** Filled with water / dilute zinc sulfate and a zinc plate.
- **Jumper Across the Partition:** A thick copper strap or wire was soldered to the copper plate in Cell 1, arched **over the solid partition wall**, and connected to the zinc plate in Cell 2.



2. The Fuller Trough Design (The Sub-Divided Trough)

In this variant, the trough had **two layers of dividers**:

- **Alternating Dividers:** The trough used alternating **solid slate dividers** (to isolate cells electrically) and **unglazed porous earthenware tiles** (to act as the ionic diaphragm between the two halves of a single cell).
- **Two Halves per Cell:** Each cell had a dry slate wall on the left and right, and a porous clay plate cemented down the middle.
 - Side A: Zinc plate in water/acid.
 - Side B: Copper plate in CuSO_4 .
- **Top Connection:** As before, the electrical connection was an external metal bridge hopping over the solid slate barrier into the next compartment.

Why the Trough Format Won Out for Telegraphy

While individual round glass or copper jars were used in small laboratories, telegraph relay stations needed **50 to 100+ cells in series**. Putting 10 cells in a single 24-inch wooden trough gave

operators:

1. **A compact, stackable footprint** that fit onto multi-tier battery racks.
2. **Reduced wire clutter**, since adjacent cells were linked by short rigid straps right over the partition rim.
3. **Spill containment**, with the outer wooden trough coated in bitumen/pitch to prevent leaks.

Sources

1. [Electrical telegraphy in the United Kingdom - Wikipedia](#)
-

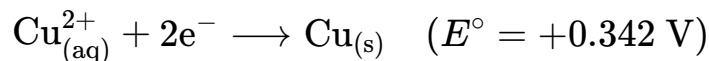
User prompt: Is the copper sheet in the copper sulfate just a current collector? Or is it involved chemically

Response: It is actively involved in the electrochemistry as the deposition substrate, not just an inert current collector.

In a Daniell cell, the copper sheet serves as both the electrical lead and the physical destination where copper ions plate out into new solid metal.

What Actually Happens at the Copper Electrode

During discharge, electrons arrive at the copper cathode from the external circuit. Dissolved copper(II) ions in the electrolyte absorb those electrons directly at the metal surface:



- **Mass Gain:** Fresh metallic copper constantly plates onto the sheet. As the cell runs, the copper plate grows thicker and heavier.
- **Potential Definition:** The chemical identity of the metal and ion pair ($\text{Cu}^{2+}/\text{Cu}^0$) directly establishes the standard reduction potential of the cathode half-cell (+0.34 V vs. SHE).

Could You Use an Inert Material Instead?

- **Graphite or Carbon:** If you replaced the copper sheet with a graphite plate or carbon rod, the cell would still work at the same potential (~1.1 V). The Cu^{2+} ions in the CuSO_4 solution would simply electroplate a thin layer of copper directly onto the carbon surface within seconds of closing the circuit, effectively turning it into a copper electrode in situ.
 - **Why Copper Sheet Was Used:** Starting with clean copper provides zero nucleation overpotential, excellent electrical conductivity, and prevents the flaking or adhesion issues that happen when copper plates onto inert graphite over long discharge periods.
-

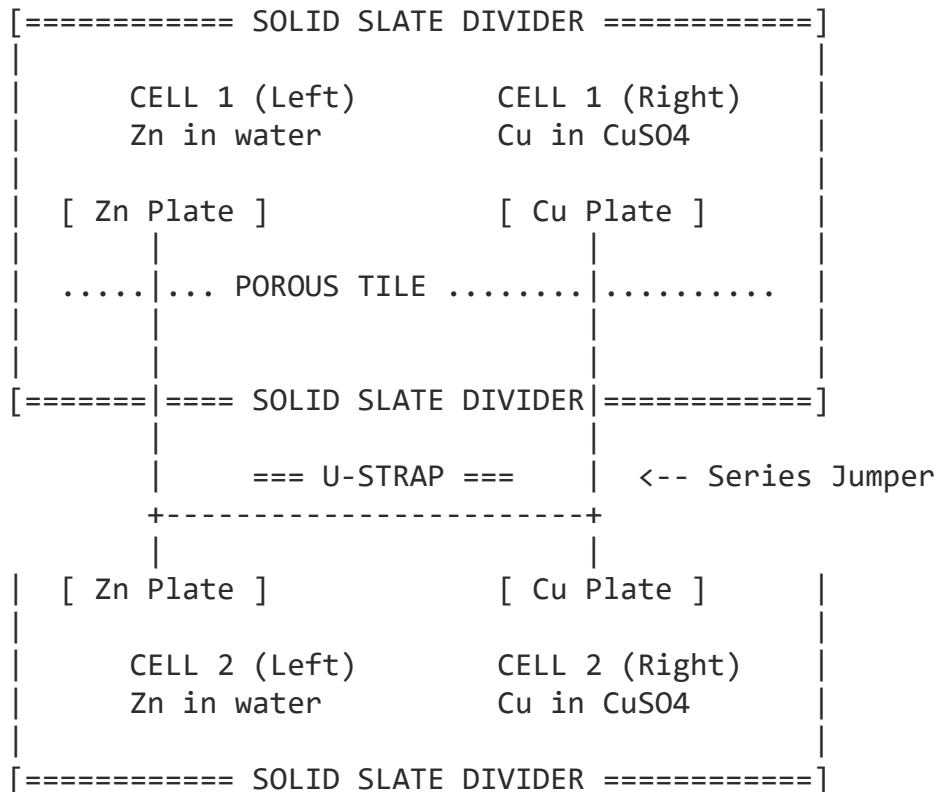
User prompt: The muirhead design was connected from the zinc rod to the copper sheet in the next cell. The fuller design would be a simple strap from the zinc side to the copper side.

Response: **Exactly.**

In both designs, the jumper bridges across the **solid insulating partition** to put the cells in series:

- **Muirhead Layout:** A long jumper wire or formed strap extends from the center **zinc rod** inside the nested porous cup, arches up and out of the pot, crosses the solid cell bulkhead, and connects to the large **copper sheet** lining the outer chamber of the adjacent cell.
- **Fuller Layout:** Because the porous tile divides the compartment into two side-by-side halves, the plates sit right next to the outer walls. A short, rigid metal "U-strap" or saddle clamp simply hops over the top edge of the **solid slate divider**—clamping the zinc plate in Cell 1 to the copper plate in Cell 2.

--- FULLER SUB-DIVIDED TROUGH (TOP VIEW) ---



Why the Fuller Design Was Popular in Shops

The Fuller setup mirrored the look and feel of the original **Cruickshank layout**, but with the critical operational upgrade:

1. **Zero through-hole leaks:** The solid bulkheads remained unbroken and completely sealed with pitch/cement.
2. **Simplified plate replacement:** When a zinc plate dissolved or fouled, an attendant could unclip the top U-strap and slide the old zinc out of its slot without disturbing the cement seal or draining the adjacent copper chamber.

User prompt: The cruickshank layout didn't have holes in the bulkheads? Or do you mean leaks around the edges of the bipolar plates? Wouldn't that also be a problem sealing the slate or ceramic?

Response: Yes, exactly—the leaks were around the perimeter edges where the plates met the wooden groove, not through holes in the metal.

In Cruickshank's original design, the metal plates themselves *were* the bulkheads.

Why Sealing Slate/Ceramic in Pitch Was Far More Reliable

Sealing a non-metal partition (slate, glazed porcelain, earthenware tile) with rosin/pitch cement proved dramatically more dependable over time than sealing a metal plate, due to three physical mechanisms:

- **1. Thermal Expansion Matching:**

- **Pine Rosin / Pitch:** Thermal expansion coefficient $\alpha \approx 60\text{--}80 \times 10^{-6} / \text{K}$.
- **Slate / Terracotta Tile:** $\alpha \approx 8\text{--}12 \times 10^{-6} / \text{K}$.
- **Wood (along/across grain):** $\alpha \approx 5\text{--}30 \times 10^{-6} / \text{K}$.
- **Zinc & Copper:** $\alpha \approx 17\text{--}30 \times 10^{-6} / \text{K}$. More importantly, as copper and zinc conducted heat from chemical reactions or current, the metal expanded and contracted rapidly. The slick, non-porous metal faces sheared away from the rigid resin cement, forming hairline boundary cracks that leaked electrolyte across cells.

- **2. Porous Mechanical Keying:**

- Ground slate, rough unglazed porcelain edges, and terracotta tiles have microscopic surface porosity and tooth.
- Hot molten rosin and pitch wick directly into the ceramic surface grain before solidifying, creating a deep, locked mechanical anchor. Pitch bonds tenaciously to stone and wood, whereas its bond to smooth, oxidized zinc or copper sheet is prone to delamination.

- **3. Galvanic Creep & Chemical Undermining at the Seal Edge:**

- In a Cruickshank trough, active battery acid sat right against the joint where the zinc met the resin. As the zinc dissolved during discharge, the metal surface *under* the cement physically receded, undermining the joint.
- In a slate-divided trough, the slate partition is completely inert—acid and salts cannot dissolve it, so the cement-to-slate interface remains undisturbed indefinitely.

By separating the **hydraulic partition wall** (inert slate/tile permanently sealed in pitch) from the **replaceable electrodes** (slid in loosely and connected over the top rim), telegraph engineers solved the constant weeping and inter-cell short circuits that plagued Cruickshank troughs.

User prompt: I can find 4" ceramic and slate tiles, that would make the replica a bit smaller. I could cast larger plaster of Paris separators. Slate is harder, unless it is possible to cut pieces out of slate cheeseboards 8"x12"

Response: Yes, slate cheeseboards cut surprisingly easily—they are actually one of the cheapest and most authentic sources of genuine raw slate for workshop projects.

Because slate is a fine-grained, foliated metamorphic rock (essentially compressed clay and silt), it has a Mohs hardness of only **2.5 to 4.0**. It behaves much softer than granite or vitreous porcelain, but it is brittle and prone to delaminating along its natural cleavage planes if forced.

How to Cut Slate Cheeseboards Cleanly

- **Best Method: Diamond Wheel on an Angle Grinder (or Wet Tile Saw)**
 - A standard continuous-rim or fine-serrated diamond masonry blade (4" or 4.5") on an angle grinder cuts 5–8 mm thick cheeseboard slate in under 30 seconds.
 - Clamp a rigid straight-edge (scrap wood or steel ruler) to the slate as a guide.
 - **Cut all the way through in 2–3 light passes** rather than scoring and snapping. Snapping along a score line can cause the internal natural grain to split diagonally across layers.
- **Alternative: Carbide/Diamond Grit Jigsaw Blade or Hacksaw**
 - A fine grit jigsaw blade run at slow speed without orbital action will cut straight lines through cheeseboard slate.
- **Dust & Safety:**
 - Slate dust contains crystalline silica. Cut outdoors and use a continuous fine spray/dribble of water from a squirt bottle to keep dust suppressed and blade temps cool.
- **Edge Dressing:**
 - Once cut, the edges can be trued up or deburred in seconds with 80-grit to 120-grit wet sandpaper on a flat block.

Casting Plaster of Paris Separators

For the unglazed porous partitions, casting your own **Plaster of Paris (hemihydrate gypsum)** tiles is straightforward and historically sound (gypsum diaphragms were standard DIY substitutes in 19th-century chemistry labs):

- **Thickness:** Aim for **4 mm to 6 mm** (~3/16" to 1/4"). Any thinner and they become fragile when wet; much thicker and internal ionic resistance rises.
- **Mix Ratio:** Mix standard plaster slightly "wet" (approx. **100 g plaster to 70–75 g water**) to increase the open capillary porosity of the cured gypsum crystal matrix without compromising

structural handling.

- **Release Mold:** Pour into an open silicone baking mold, a 3D-printed form lined with parchment/PET tape, or Lego bricks on a smooth sheet.
- **Curing & Pre-wetting:** Let cure for 24 hours. **Always pre-soak the dry plaster tiles in clean water or dilute electrolyte before seating and running the cell** so air bubbles inside the micro-pores are fully displaced.

Sources

- 1. [Tips on Cutting and Preparing Slate - Second Use Building Materials](#)
- 2. [Working with Slate - Lenco Heaven](#)

User prompt: I think I'd make a 3dp printed mold for a silicone mold for the plaster. What kind of current was needed to run a telegraph line ?

Response: **Typical telegraph lines operated on surprisingly tiny currents: roughly 15 to 50 milliamperes (0.015 to 0.050 A).**

Because the iron telegraph wire stretched for tens or hundreds of miles, lines were high-resistance, voltage-limited circuits rather than high-current systems.

Standard Operating Currents by Equipment Type

Instrument / System	Operating Current	Why This Range Was Chosen
Sensitive Polarized Relays (Long-distance trunk lines)	15 to 25 mA	High-turn electromagnets could trip a sensitive, counterbalanced armature on minimal current.
Standard Morse Registers & Sounders (Local loops / Main lines)	30 to 50 mA	Needed enough magnetic force to make a crisp, loud physical "click" on brass stops without an intermediate relay.
Needle Telegraphs (British Railways / Cooke & Wheatstone)	10 to 25 mA	Moving a lightweight magnetic needle over a dial required barely any power.

Why Voltage, Not Current, Dictated Battery Size

A 100-mile single iron wire line (plus ground return, key, and electromagnet coil resistance) easily totaled 1, 500 to 3, 000 Ω of circuit loop resistance.

By Ohm's Law ($I = V/R$):

- To push just 30 mA (0.030 A) through a 2, 000 Ω line:

$$V = I \times R = 0.030 \text{ A} \times 2000 \Omega = 60 \text{ Volts}$$

- Since a single Daniell cell delivers $\approx 1.1 \text{ V}$, telegraph stations had to string **50 to 60 Daniell cells in series** (often five or six 10-cell wooden troughs ganged together) just to overcome the line resistance.

Because each cell only needed to deliver $\sim 30 \text{ mA}$, small 3" or 4" plates operated at a very comfortable current density ($< 0.5 \text{ mA/cm}^2$). This kept polarization negligible and allowed a battery bank to run reliably for weeks before needing water dilution or a top-up of blue crystals.

User prompt: So the key was used to send and the sounder was the receiver?

Response: Yes, exactly—the telegraph key was the momentary switch used to transmit, and the sounder was the acoustic receiver.

Every station along the line had both a key and a sounder wired in series with the battery and the single overhead iron line wire (using the earth as the return path).

How the Two Instruments Worked Together

- **The Key (Transmitter):**
 - A spring-loaded, single-pole momentary switch.
 - Tapping the knob brought an upper contact down onto an insulated lower anvil, closing the electrical loop and sending current down the line for the duration of the tap (short for dots, longer for dashes).
 - **The "Circuit Closer" Switch:** Notice the small black lever on the left of the key above. American Morse systems ran on a **normally closed loop**. When an operator finished sending, they closed this lever to complete the circuit for other stations. When ready to send, they opened the lever, taking control of the line.
- **The Sounder (Receiver):**
 - An electromagnet with a spring-loaded iron armature attached to a heavy brass crossbar.
 - When current flowed through the coils, the electromagnet pulled the armature down against an adjustable brass anvil with a sharp **"click"** (down-stroke).
 - When current stopped, a return spring yanked the crossbar up against an upper stop screw with a slightly different **"clack"** (up-stroke).
 - Operators didn't listen to continuous tones—they read Morse code purely by the **rhythm and time interval between the down-click and the up-clack**.

Long-Distance Lines: Adding the Relay

On long lines (50–200+ miles), line resistance weakened the signal down to ~15–20 mA—too weak to pull a heavy sounder crossbar with a loud click.

Telegraphers placed a sensitive **Relay** on the main line:

1. The weak line current tripped the featherweight armature of the relay.
2. The relay contacts acted as an automated key for a separate, private "**Local Circuit**" in the office.
3. A small local 1- or 2-cell Daniell battery then powered the loud, heavy **sounder** directly at full current (50–100 mA).

User prompt: Were there repeater stations automatically amplifying the signal

Response: **Yes, automatic telegraph repeaters were invented in the 1850s and 1860s specifically to solve the distance limit without human re-transcription.**

Before repeaters, a telegraph operator at an intermediate station (every 150–250 miles) had to listen to the incoming Morse sounder with one hand and manually re-tap the message onto the next line with their own key.

The Engineering Dilemma: The Feedback Loop

Building an automatic repeater was not as simple as wiring Relay 1 directly into Line 2:

- If **Line East** activates **Relay East**, which keys **Line West**...
- The moment **Line West** activates, it trips **Relay West**, which immediately keys **Line East** right back.
- This creates a permanent electromechanical feedback loop (called "lock-up" or "chatter").

Because telegraph lines were **bidirectional** (either station could send or interrupt), an automatic repeater had to detect which direction the message was traveling and instantly lock out the reverse path.

How 19th-Century Automatic Repeaters Worked

Inventors like **George F. Milliken**, **Thomas Edison**, and **Moses G. Farmer** developed ingenious interlocking electromechanical circuits (such as the *Milliken Repeater* and *Weiny-Phillips Repeater*):

- **Dual Main Line Relays:** The station had two separate line circuits—one going East, one going West—each with its own large battery bank.
- **Mechanical / Auxiliary Lockout Relays:** When Line East pulsed, Relay East tripped an intermediate local transmitter. As this transmitter closed the circuit for Line West, it *simultaneously* energized a holding coil or dropped a mechanical latch onto Relay West.
- **Breaking the Feedback:** This physical lockout prevented Relay West's contacts from opening when Line West opened, stopping the feedback loop while faithfully repeating the

pulses forward at full line voltage.

What It Achieved

By placing automatic repeaters in junction towns every 200 to 300 miles, networks like Western Union could link **New York to San Francisco (over 3,000 miles)** in a single instantaneous circuit.

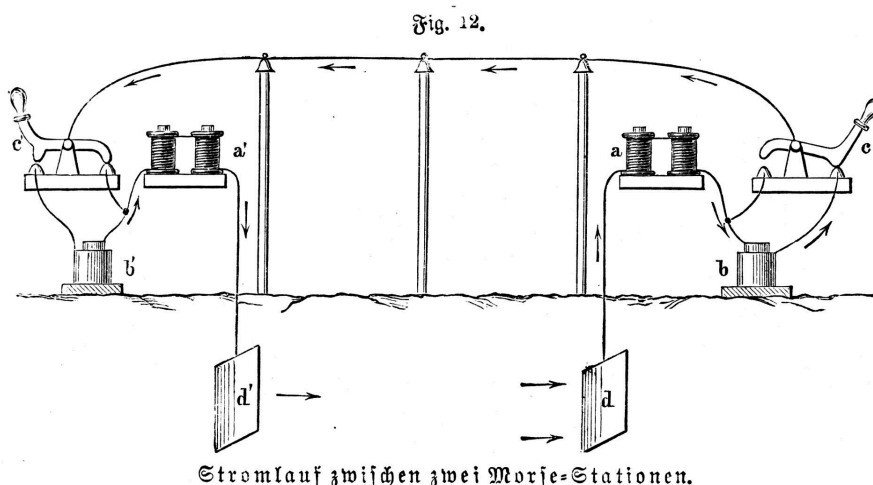
When an operator in New York tapped their brass key, a cascade of 10 to 15 automatic repeaters across the continent opened and closed in microsecond synchronization, keying fresh Daniell/gravity battery banks down each succeeding leg of wire.

Sources

1. [Thomas Alva Edison | Encyclopedia.com](#)

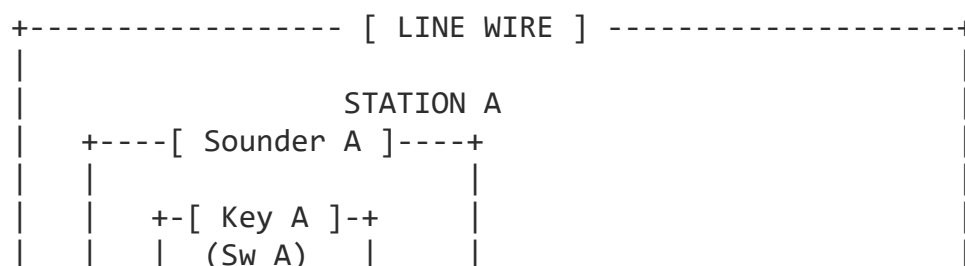
User prompt: How do I wire a two-station Morse telegraph circuit with keys, sounders, and batteries?

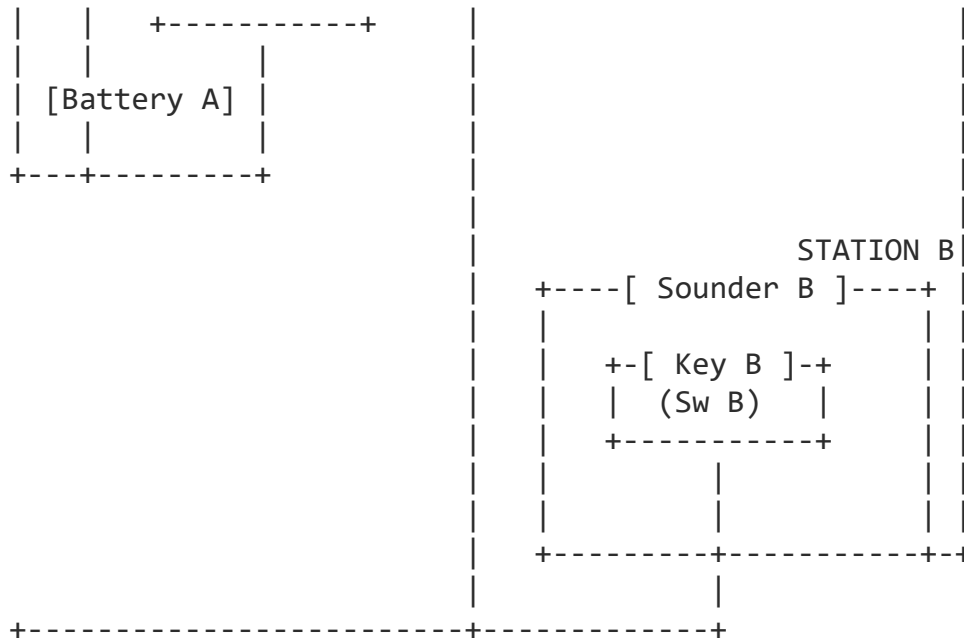
Response: The standard American system used a **closed-circuit loop**: every key, sounder, and battery along the line was wired in one continuous series circle.



The Series Wiring Schematic

On a bench or between two rooms, wire the components in a single continuous loop (using a return wire instead of an earth ground rod):





- **Station A:**

1. Battery positive (+) connects to **Line Wire** running to Station B.
2. Battery negative (−) connects to one terminal of **Key A**.
3. The other terminal of **Key A** connects to **Sunder A**.
4. The second terminal of **Sunder A** connects to the **Return Wire**.

- **Station B:**

1. The **Line Wire** connects to **Sunder B**.
2. The other terminal of **Sunder B** connects to **Key B**.
3. The other terminal of **Key B** connects back to the **Return Wire**. *(Note: A single battery bank at Station A can power the entire small loop, or you can split cells between Station A and Station B, ensuring all batteries face the same series direction: + to −).*

How to Operate the Circuit (The "Circuit Closer" Routine)

1. **Idle State (Closed Loop):**

- Both operators keep the small side lever (**circuit closer**) on their keys closed.
- Current flows continuously through the entire loop ($\approx 30\text{--}50\text{ mA}$).
- Both sounders remain energized and pulled down in the quiet "clack" position.

2. **Taking the Line to Transmit:**

- Station A flips open their side lever.

- The circuit opens: current drops to zero, and **both sounders pop up**.

3. Sending Morse:

- Station A taps the key knob.
- Every tap completes the circuit, pulling **both Sounder A and Sounder B down simultaneously** with a loud "click."
- Station A hears their own transmission (local sidetone), while Station B reads the message.

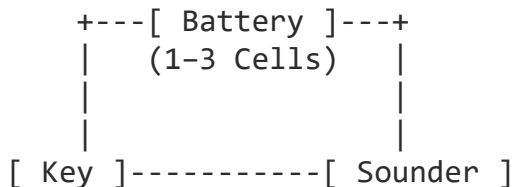
4. Restoring the Line:

- Once finished, Station A flips their side lever shut.
- Current resumes, keeping the line closed and ready for Station B to open their lever and reply.

User prompt: So just a sounder and a key wired to the battery would be enough to simulate the sending side of the loop.

Response: Yes, exactly—wiring a single key, sounder, and battery in series gives you a complete, authentic "local practice set."

Historically, telegraph companies sold this exact setup as a standalone learner's set mounted on a single walnut board for trainee operators to practice keying and reading Morse code.



How to Hook It Up on the Bench

1. The Circuit:

- Wire the (+) **terminal** of your battery trough to one post of the **Key**.
- Wire the other post of the **Key** to one coil terminal of the **Sounder**.
- Wire the second coil terminal of the **Sounder** back to the (−) **terminal** of the battery.

2. Key Setup:

- Leave the key's side switch (circuit closer) **open** so current only flows when you press the knob.

- Every time you tap a dot or dash, the circuit completes, energizing the electromagnet to slam the armature down against the brass anvil with a crisp "**click**"; releasing it lets the spring yank the crossbar up with a "**clack**."

Voltage & Battery Sizing for a Local Set

Because there are no miles of high-resistance line wire on a workbench, you only need to overcome the internal resistance of the sounder coil itself:

- **For a Low-Resistance Sounder ($4\ \Omega$ – "Local Sounder"):**
 - Needs roughly 100 to 250 mA for a loud strike.
 - 1 to 2 Daniell cells (≈ 1.1 to 2.2 V) is plenty.
 - **For a High-Resistance Mainline Sounder ($20\ \Omega$ to $150\ \Omega$):**
 - Needs roughly 30 to 50 mA.
 - 2 to 4 Daniell cells (≈ 2.2 to 4.4 V) provides enough drive voltage.
-

User prompt: Why were the troughs in England made of mahogany or teak? Wouldn't oak have been cheaper? Were these mitered boxes or some other joinery?

Response: **Why Mahogany and Teak Beat Oak**

While English oak was locally available, it had fatal chemical and physical flaws for battery enclosures:

- **Tannic Acid Attack:** Oak contains very high concentrations of natural tannic acid. When exposed to warm, humid environments or sulfuric acid vapors, the tannins leach out, corrode nearby metals, and chemically degrade the adhesion of rosin/pitch electrical cements.
- **Dimensional Instability (Warping & Movement):** Oak expands and shrinks dramatically across its grain with changes in relative humidity. A trough holding wet electrolytes would swell on the inside and dry on the outside, breaking the brittle rosin/pitch seals around the divider slots and causing cross-cell leaks.
- **Why Mahogany:** Honduran/Cuban mahogany has exceptionally low volumetric shrinkage, straight interlocking grain that resists cupping, and virtually no reactive tannins. It was the standard for Victorian scientific instrument cases.
- **Why Teak:** Teak (imported widely via the Royal Navy and British East India trade) is naturally saturated with silica and hydrophobic natural tectoquinones/oils. It is virtually rot-proof, completely impervious to acid vapors, and will not warp even under direct, continuous moisture.

Historic Joinery Methods

Simple miter joints were never used because end-grain glue bonds cannot resist the hydraulic pressure of fluid or the mechanical stress of swelling partitions. Instrument makers used heavy, interlocking mechanical joints:

- **Through Dovetails or Half-Blind Dovetails (The Standard):**

- The four outer walls of high-end lab troughs and Post Office telegraph boxes were joined with fine hand-cut **through dovetails**.
- Dovetails provided immense tensile strength at the corners, preventing the long side walls from bowing outward under the weight of the plates and acid.

- **Rabbet (Rebate) & Box Joints (Utility Telegraph Troughs):**

- Mass-produced railroad troughs often used pinned **rabbet joints** or machine-cut **finger/box joints**, frequently reinforced with brass or copper pins (iron screws/nails corroded instantly in acid fumes).

- **The Bottom Floor Joint:**

- The bottom board was not simply nailed to the bottom rim. It was seated into a continuous **plowed groove (dado/rebate)** cut around the inner base of the four walls, then completely flooded and potted with molten pitch/rosin to create a seamless, liquid-tight basin.

User prompt: What would the bottom of the trough be made of, I assume they didn't have plywood?

Response: **It was made of a single thick, solid hardwood board (usually quarter-sawn).**

Modern cross-laminated waterproof plywood did not exist in the 19th century. Instead, instrument makers and telegraph workshops used solid timber slabs with specific woodworking techniques to manage moisture and hydraulic pressure.

How 19th-Century Makers Built the Bottom

- **Solid Slab Construction:**

- The bottom was a single piece of **0.75" to 1.0" (19 mm to 25 mm)** thick solid mahogany, teak, or well-seasoned pitch pine.
- If a wide trough exceeded the width of a single available board, pieces were joined with **rubbed hide-glue joints reinforced with internal splines or tongue-and-groove milling**, but narrow single boards (4" to 6" wide) were preferred to eliminate seams entirely.

- **Quarter-Sawn Grain Selection:**

- Makers deliberately chose quarter-sawn stock (growth rings perpendicular to the face).
- Quarter-sawn boards expand and contract primarily in thickness rather than width, and they **do not cup or bow** when wet on one side and dry on the other.

- **How It Was Joined to the Sides:**

- **Captured Bottom Groove (Dado):** The bottom board was captured in a 1/4" to 3/8" deep rebate/dado plowed along the inner perimeter of the four side walls before the dovetail corners were assembled.
- **Floated & Bedded:** The board was cut slightly shy of the groove bottom to allow for minor cross-grain swelling, and the entire groove was flooded with hot **pine pitch/rosin cement** during assembly rather than rigid glue.

The Interior Waterproof "Basin"

Because no raw solid wood floor could stay completely watertight against standing acid and copper sulfate solutions indefinitely, the wooden box acted purely as a **structural exoskeleton**:

1. **Hot Pitch Flooding:** Once the box was assembled and dovetails pinned, boiling-hot pine pitch, bitumen, or rosin/wax cement was poured into the bottom to a depth of **1/4" to 1/2" (6 mm to 12 mm)**.
2. **Seamless Pan:** This created a monolithic, seamless, chemical-proof resin "pan" across the floor that permanently sealed the joints between the bottom board and side walls.
3. **Groove Seating:** The vertical divider kerfs (slots) cut into the walls terminated right into this pitch bed, ensuring no liquid could track underneath the partitions.

User prompt: Rebate or dado is a groove or right on the bottom edge.

Response: A dado is an enclosed channel spaced in from the edge, while a rebate (rabbet) is an open step cut right on the edge.

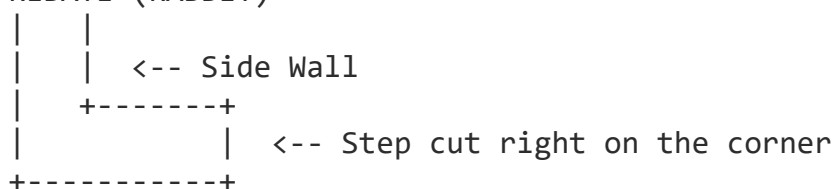
In traditional box making and trough construction, the distinction determines whether the bottom board is fully captured or recessed into the perimeter rim.

The Two Joint Profiles

- **Rebate / Rabbet (Step on the edge):**

- Cut right along the corner/edge of the board (has **one wall and an open shoulder**).
- In trough building, running a rebate around the bottom inside rim lets the floor board drop in from above (or push up from underneath), sitting flush with the bottom edge.

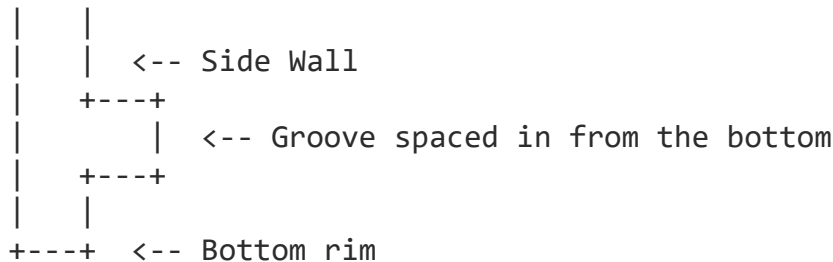
REBATE (RABBET)



- **Groove / Dado (Enclosed channel):**

- Spaced up from the edge (has **two parallel walls and an enclosed floor**).
- Technically: a *groove* runs with the grain; a *dado* (or *housing*) runs across the grain.
- In trough construction, running a groove about 1/4" to 3/8" above the bottom edge lets the floor board slide in and get **fully captured** on all four sides like a drawer bottom before the final dovetail or corner joint is closed.

CAPTURED GROOVE



Which Was Used for Troughs?

- **Captured Groove (Most Common for Quality Troughs):**
 - Machining a 1/4" wide groove spaced 3/8" up from the bottom edge locks the hardwood floor mechanically into the side walls. Even under hydraulic weight, the bottom cannot drop out or sag.
- **Bottom Edge Rebate (Utility Telegraph Troughs):**
 - A 3/8" deep rebate cut around the bottom inner perimeter allowed the floor to be dropped into the assembled frame, pinned horizontally with brass dowels, and then flooded with molten rosin/pitch to seal the corner seam.

User prompt: 19 mm is pretty thick?

Response: Yes, 19 mm (~3/4") is beefy for a small box, but it was necessary to withstand the physical demands of 19th-century battery use.

For a standard jewelry or tool box, 19 mm would be overkill. For a wet battery trough, that thickness solved three structural and operational issues:

Why 19 mm Was the Historic Standard

- **1. Heavy Dado Depth vs. Remaining Wall Strength:**
 - To hold 10 to 50 heavy divider plates upright against fluid movement, builders cut vertical kerfs/dados **3 mm to 5 mm (1/8" to 3/16") deep** into the inner sidewalls.
 - In a thin 10 mm (~3/8") board, a 4 mm groove would remove almost half the wall thickness across 20+ vertical lines, effectively turning the sides into flexible, perforated scoreboards prone to snapping.

- In a **19 mm wall**, a 4 mm groove leaves **15 mm (~5/8") of solid hardwood** behind the slot, keeping the long box rigid and straight.
- **2. Preventing Cross-Grain Cupping & Bowing:**
 - The inside of the trough was constantly saturated with wet electrolyte, while the outside was exposed to dry air.
 - This extreme humidity differential creates severe asymmetric expansion that causes thin boards to curl and cup outward. A 19 mm quarter-sawn board provides the mechanical stiffness needed to resist bowing, keeping the cement seals around the plates intact.
- **3. Fluid Mass and Handling:**
 - A full 10-cell trough loaded with slate partitions, zinc plates, copper sheets, crystals, and liquid easily weighed **15 to 25 lbs (7 to 11 kg)**.
 - The trough had to be carried by its ends or rim during maintenance without flexing; flex would instantly shatter the brittle rosin/pitch cement seams and cause inter-cell cross-leaks.

Modern Scale-Down for a DIY Bench Model

For a smaller 5- to 10-cell decorative or experimental bench replica using 3" or 4" plates:

Part	Historic Standard	Practical Modern DIY
Side & End Walls	19 mm to 25 mm (3/4" to 1")	12 mm to 15 mm (1/2" to 5/8")
Bottom Board	19 mm (3/4")	12 mm (1/2") solid hardwood
Kerf / Groove Depth	4 mm to 5 mm	2.5 mm to 3 mm (approx. blade kerf)

Dropping down to **12 mm (~1/2")** keeps the box lighter and more proportioned for a small desk set while still leaving 9 mm of meat behind standard saw-blade kerfs.

User prompt: I'm thinking I need to practice with some cheaper wood first. Any suggestions for similar machining characteristics but cheaper? Maybe without the good movement and moisture characteristics

Response: Here are the best practice woods that mimic genuine mahogany’s crisp machining and chisel behavior without the high price tag:

Top Practice Species

- **1. Yellow Poplar (*Liriodendron tulipifera*) — The Best Budget Stand-In**
 - **Cost:** Very cheap; widely available at local home centers and hardwood yards.

- **Machining Behavior:** Machines smoothly without dulling blades, takes clean table-saw dados, and routes cleanly with minimal burning.
- **How it compares:** It is slightly softer than mahogany (Janka hardness ~540 vs. ~800–900), which means hand chisels cut through it with less resistance, but end-grain can get slightly "fuzzy" if tools aren't razor sharp. It moves significantly with moisture changes, making it ideal purely for sizing, setup, and joint trials.
- **2. Sapele or African Mahogany (*Khaya*) — The Closest Direct Match**
 - **Cost:** Roughly 40–50% cheaper than genuine Honduran mahogany (*Swietenia macrophylla*).
 - **Machining Behavior:** Shares mahogany's medium-dense feel under chisels and router bits, producing sharp, well-defined shoulders in dados and dovetails.
 - **How it compares:** Sapele is slightly harder and can have ribbon-stripe interlocked grain that tears out if planed against the grain, but for sawing, dadoing, and slot cutting, it feels almost identical to genuine mahogany.
- **3. Soft Maple (Red Maple / Silver Maple)**
 - **Cost:** Moderately cheap, readily stocked at hardwood suppliers.
 - **Machining Behavior:** Holds ultra-crisp edges in narrow kerfs without splintering or tear-out.
 - **How it compares:** Slightly harder than mahogany, but its uniform closed grain lets you practice delicate joinery, narrow divider slots, and dovetails with high precision.
- **4. Clear Radiata or White Pine (Skip construction-grade yellow pine)**
 - **Cost:** Inexpensive.
 - **Machining Behavior:** Fast to cut, but soft.
 - **How it compares:** The pronounced difference between soft springwood and hard latewood can cause hand chisels to wander or tear out across grain lines. Useful for quick gross-dimension proof-of-concepts, but less forgiving for fine, tight kerfs.

Recommended Approach

1. **Poplar or scrap pine for the first dry run:** Use 1/2" or 3/4" poplar to dial in table saw fence stops, check divider slot spacing against your cast plaster tiles, and test corner joinery.
2. **African Mahogany / Sapele for the finished build:** If you want the authentic look and pitch-resistant stability of mahogany without the genuine CITES-regulated price, African mahogany or Sapele offers an excellent balance of cost, workability, and durability.

Sources

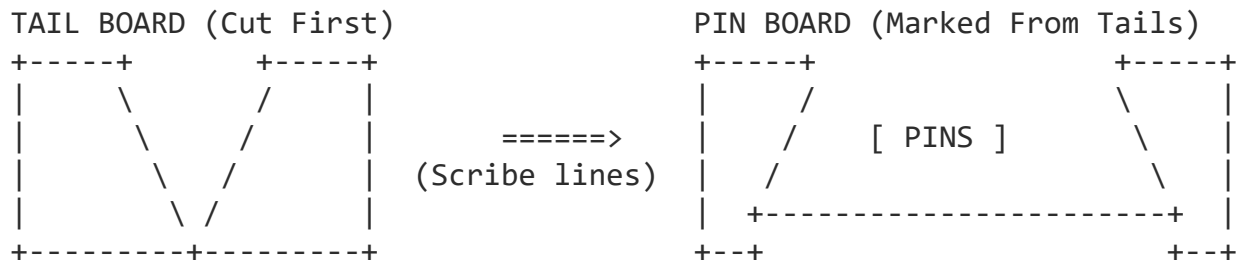
1. [Is poplar a good body wood? : r/Luthier - Reddit](#)
2. [Mahogany Wood Comparison of Sapele, African Mahogany, Sipo](#)

User prompt: Poplar it is! I think sizing a 12" long trough for practice joinery holding 4" bipolar sheets. I really want to try dovetails but they terrify me

Response: Dovetails have an intimidating reputation, but on a straight box with four corners, they are simply a series of straight saw cuts and light chisel work. Poplar is forgiving to learn on: it pares cleanly across the grain without tearing out like pine or dulling your edges like rock maple.

The Secret to Stress-Free Dovetails: "Tails First"

Cutting the **tail board** (usually the long sides) first gives you a physical, 1:1 marking template to scribe the pins directly onto the mating pin board. You never have to measure angles or pin spacing twice.



A 5-Step Beginner Workflow

1. Mark the Baseline Accurately:

- Set a wheel marking gauge to the *exact* thickness of your mating board (or a hair thicker, so pins end up proud for flushing later).
- Scribe this depth line all the way around all four ends of every board. **Never cut past this line.**

2. Lay Out Simple, Wide Tails:

- Don't make microscopic, violin-maker pins on your first box. For a ~4.5" high trough, use **2 or 3 wide tails**.
- A classic **1:6 or 1:8 slope** (roughly 7° to 9°) is standard. You don't need a fancy gauge—draw a 1" offset over 6" on a scrap card to set a sliding bevel.

3. Saw to the Waste Side of the Line:

- Clamp the tail board vertically in the vise.
- Make your angled cuts down to the baseline. Keep the saw blade strictly on the **waste side** of your pencil/knife line.

4. Clear the Waste:

- Use a coping saw or fret saw to remove the bulk of the waste between the tails in one quick U-shaped cut just above the baseline.

- Place a sharp bench chisel directly in your scribed baseline and pare flat to clean the bottom floor.

5. Transfer to the Pin Board:

- Clamp the pin board vertically in the vise. Lay the cut tail board flat on top, aligning the edges flush.
- Use a sharp marking knife or razor blade to trace the shape of the tails directly onto the end-grain of the pin board.
- Saw straight down on the **waste side** of those knife lines, clear the center with your coping saw/chisel, and test fit.

Three "No-Panic" Practice Tips

- **Leave Them Proud:** Make your baseline ~0.5 mm deeper than the board thickness. When assembled, the pins and tails will stick out slightly. Five seconds with a hand plane or 120-grit sanding block flushes them down dead-flat and makes the joint look seamless.
- **Blue Tape Trick:** If seeing the lines on poplar end-grain is tough, put a strip of blue painter's tape across the end-grain, scribe your knife lines through it, and peel away the waste sections. Saw right along the crisp blue tape edge.
- **Saw a Scrap First:** Take two 4" offcuts of poplar and cut just **one single tail and pin** to get the muscle memory down before touching your 12" trough sides.

User prompt: I think the sides need to be at least 5” high to accommodate the bottom 1/2” plus a 1/4” rim, and maybe a 1/4” deeper than the plates at the top? Printable template please

Response: Here is a visual dovetail layout generator tailored to your dimensions. You can adjust the parameters, toggle guides, and hit **Print 1:1 Scale Template** to generate full-scale templates to glue directly onto your poplar test pieces.

Vertical Stack Breakdown (5.0" Total Height)

Your sizing math works out cleanly to a standard **5-inch finished board height**:

Section	Height	Purpose / Function
Top Air Gap	0.250" (1/4")	Freeboard to prevent electrolyte sloshing and give room for jumper clips.
Active Plate Zone	4.000" (4")	Full immersion height for your 4" square plates and cast plaster tiles.
Pitch Floor Bed	0.250" (1/4")	Monolithic layer of poured rosin/pitch cement sealing the floor seam.

Section	Height	Purpose / Function
Captured Groove	0.250" (1/4")	1/4" wide groove plowed into the side walls to capture the floor board.
Bottom Rim	0.250" (1/4")	Solid wood meat below the groove to prevent the bottom edge from splitting out.
Total Board Width	5.000" (5")	Standard S4S 1x6 nominal stock (5.5" actual) ripped down to 5.0".

Step-by-Step Template Usage

1. **Verify Printer Scale:** Print the template from the widget. Measure the printed 1-inch check bar with a steel rule before sticking it down. If it is slightly off, set your printer dialog to **"Actual Size" / 100% scale** (disable "Fit to Page").
2. **Adhere to End Grain:** Cut the paper template out and attach it to your squared poplar ends using a light mist of spray adhesive (or a standard purple glue stick—it washes off with warm water).
3. **Score the Baseline:** Score your wheel marking gauge across the baseline right through the paper into the wood before making your first saw cut.
4. **Saw the Waste:** Saw strictly in the crosshatched waste areas to leave your pencil/knife lines intact.

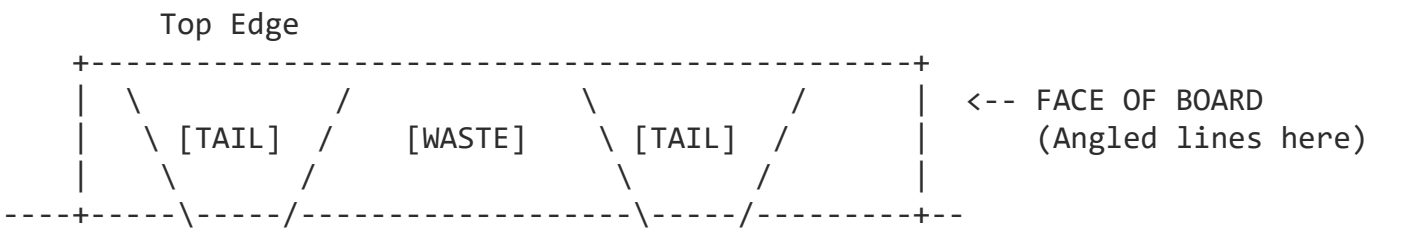
User prompt: The template for tails goes on the face of the board?

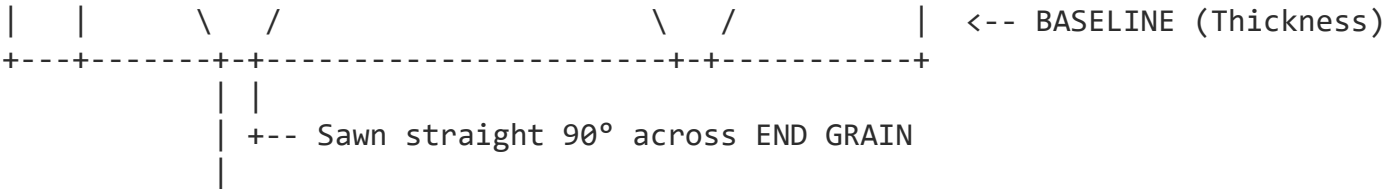
Response: **Yes, exactly—the tail template goes on the broad face of the board.**

The layout logic follows this rule:

- **Tails (Face Pattern):** The angled slopes (1 : 6) live on the **face** of the board. The cut on the end-grain is a simple 90° square line straight across the thickness.
- **Pins (End-Grain Pattern):** The angled slopes (1 : 6) live on the **end-grain** of the mating board. The cut down the face is a 90° plumb line down to the baseline.

--- TAIL BOARD LAYOUT (Side Wall) ---





How to Apply and Cut the Tail Template

1. **Align to the End:** Glue the face template flat on the wide face of your poplar board, aligning the top edge of the template flush with the cut end-grain.
2. **Square Across the End-Grain:** Use a small engineer's try-square or combo square to extend pencil lines 90° **straight across the top end-grain** from each point where the angled tail lines touch the edge.
3. **Clamp Vertically in the Vise:**

◦ Look down at the end-grain: your saw stays perfectly vertical (90°) relative to the board's thickness.

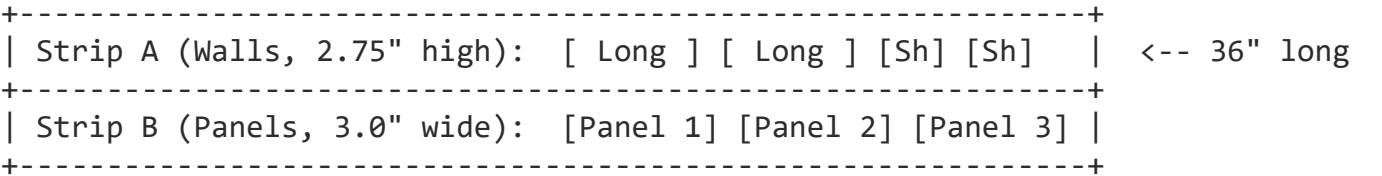
◦ Look at the face: tilt your saw left or right to track the printed 1 : 6 angled line down to the scribed baseline.
4. **Skip Printing the Pin Template:** Once the tails are cut and the waste is chiseled out, don't use a printed paper template for the pin board. Simply rest the cut tail board on the pin board's end-grain and **trace the pins directly with a sharp knife**. That guarantees a zero-tolerance match even if your saw wandered slightly on the tails.

User prompt: I have a couple of pieces of radiata pine, 3/4"x 6"x36". If I wanted to make one into a dovetailed box with a lid and bottom recessed into rabbets, and longer than wide, what are my dimensional options? Or do I need more than one piece of wood?

Response: To make the 4 walls, a lid, and a bottom, **using two boards is vastly easier and gives you full height**, but **one board is possible if you build a shallower box and resaw or edge-glue the offcuts**.

Option 1: Using ONE Single Board (Shallower Box)

A single 6" × 36" board has 216 sq inches of surface area. To get all 6 faces (4 sides + top + bottom), you must rip the board lengthwise into strips:



- **Ripping:** Rip the 6" width into **Strip A (2.75" wide)** and **Strip B (~3.0" wide)** (accounting for blade kerf).
- **The 4 Walls (from Strip A):**
 - 2 Long sides: **10.5" long × 2.75" high**
 - 2 Short sides: **6.5" long × 2.75" high**
 - Perimeter: $10.5 + 10.5 + 6.5 + 6.5 = 34.0''$ (leaving 2" for crosscut kerfs and squaring).
- **Lid and Bottom (from Strip B):**
 - Crosscut Strip B into three 11" lengths.
 - Edge-glue two pieces together to make a $6.0'' \times 11.0''$ panel for the lid (or resaw $3/4''$ stock into two $5/16''$ thin boards on a bandsaw/table saw to get both top and bottom).
- **Final Box Dimensions:**
 - **Outside:** $\approx 10.5'' \text{ L} \times 6.5'' \text{ W} \times 2.75'' \text{ H}$
 - **Inside (with $3/4''$ walls):** $\approx 9.0'' \text{ L} \times 5.0'' \text{ W} \times 1.75'' \text{ H}$

Option 2: Using TWO Boards (Full Height, Minimal Hassle)

Using two boards eliminates awkward edge-gluing and gives you the depth needed for a 5" tall battery trough, tool chest, or storage box:

- **Board 1 (All 4 Walls):**
 - Kept at full 5" to 6" width.
 - Crosscut directly into:
 - 2 Long sides: **11.5" long × 5.0"–6.0" high**
 - 2 Short sides: **6.0" long × 5.0"–6.0" high**
 - Total length: $11.5 + 11.5 + 6.0 + 6.0 = 35.0''$ (clean fit within 36").
- **Board 2 (Lid, Bottom & Test Pieces):**
 - Crosscut two 11.5" lengths at full 6" width for the **Lid** and **Bottom**.
 - Leaves a **13" scrap piece** to practice cutting dovetail joints before touching the box pieces.
- **Final Box Dimensions:**

- **Outside:** $\approx 11.5''\text{ L} \times 6.0''\text{ W} \times 5.5''\text{ H}$
- **Inside:** $\approx 10.0''\text{ L} \times 4.5''\text{ W} \times 4.0''\text{ H}$ (with 1/2" recessed floor and lid).

Comparison & Recommendation

Approach	Max Outer Dimensions ($L \times W \times H$)	Woodworking Required	Practice Scrap Left Over?
1 Board	10.5'' $\times$ 6.5'' $\times$ 2.75''	Rip cuts, edge-gluing or resawing	None (zero margin for error)
2 Boards	11.5'' $\times$ 6.0'' $\times$ 5.5''	Simple crosscuts only	13" $\times$ 6" offcut for practice joints

Recommendation: Use **two boards**. It gives you a proportioned 11.5'' $\times$ 6'' box that can hold 4" battery plates or tools, avoids edge-gluing thin panels, and gives you a 13" offcut to practice your first test dovetails.

User prompt: I’m just reading a translation of parts of Volta’s original 1880 paper. It seems like he used himself as the voltmeter?! He also describes “I dispose, therefore, a row of several basons or cups of any matter whatever, except metal, such as wood, shell, earth or rather glass (small tumblers or drinking glasses are the most convenient), half filled with pure water, or rather salt water or ley: they are made all to communicate by forming them into a sort of chain, by means of so many metallic arcs, one arm of which, [Aa], or only the extremity [A], immersed in one of the tumblers, is of copper or brass, or rather of copper plated with silver; and the other, [Z], immersed into the next tumbler, is of tin, or rather of zinc.” what is ley?

Response: “**Ley**” is the archaic 18th-century spelling of **lye**—a strongly alkaline solution historically made by leaching wood ashes with water (primarily potassium carbonate and potassium hydroxide, or potash).

Volta noted that salt water or ley worked far better than pure water because the dissolved ions dramatically lowered internal electrolyte resistance. (As an aside, his landmark letter to the Royal Society was published in **1800**, establishing the dawn of continuous electrochemistry).

Volta Literally Used Himself as the Voltmeter

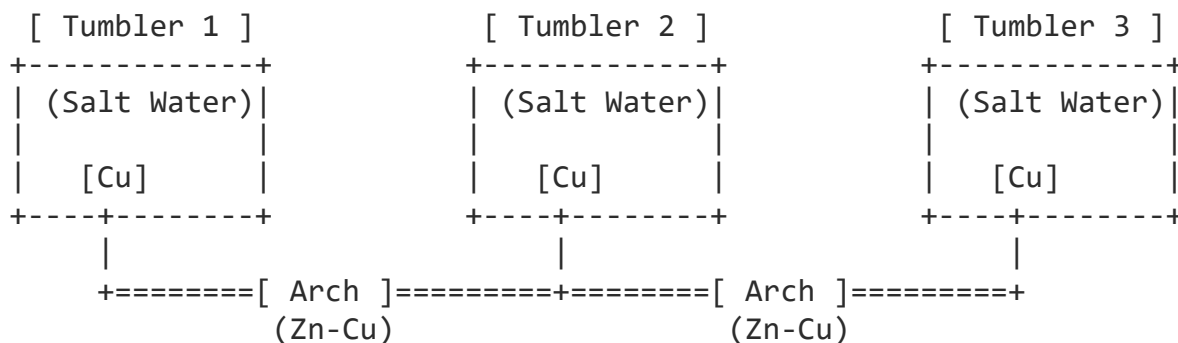
You read that right: **Alessandro Volta was his own detector and voltmeter.**

In 1800, electrical meters for continuous direct current simply did not exist. Electrostatic tools like the gold-leaf electroscope could only detect high static voltages (often requiring Volta’s own "condenser" plate to see even a faint twitch from a chemical cell). To measure current, intensity, and polarity, Volta turned to his own nervous system:

- **The Tongue (Low-Voltage Sensor):** To test 1 or 2 bimetallic couples (around 1 to 2 V, well below the threshold of skin sensation), he touched the metal ends to the tip and base of his moist tongue. He evaluated voltage by the sharpness of the **acidic/sour taste** (at the zinc anode) versus an **alkaline/soapy taste** (at the copper/silver cathode).
- **The Eyes (Visual Phosphenes):** By placing one lead in his mouth and touching the other to his eyelid or forehead, the sudden make-and-break of the circuit stimulated the optic nerve, producing a brief **flash of light** (phosphene) whose brightness indicated circuit vigor.
- **Fingers in Saline Cups (Shock Intensity / High Voltage):** For larger piles of 20, 40, or 60 pairs (20 to 60 + V), he dipped his fingers or whole hands into bowls of salt water connected to the pile terminals. He quantified the "force" of the pile by where he felt the continuous prickling shock: whether the sensation stopped at the knuckles, traveled up to the wrists, or jolted his forearms and elbows.

The "Crown of Cups" (*Couronne de Tasses*)

The passage you quoted describes Volta's **Crown of Cups**, which he introduced in that same paper as the horizontal, non-draining alternative to his vertical "Pile":



Instead of wet paper discs that dried out or squished together under the weight of a vertical pile, he bent bimetallic strips of copper/brass and zinc into **inverted U-arches** and dropped them into drinking glasses filled with brine or lye.

This horizontal layout was the direct ancestor of Cruickshank's trough battery (which fused the cups into a single slotted wooden box) and Daniell's two-fluid cell decades later.

User prompt: The attached pdf is the original Volta paper. Can you give me a translation?

Response: An English translation of Alessandro Volta's landmark letter to Sir Joseph Banks, published in the Philosophical Transactions of the Royal Society of London (1800):

XVII. On the Electricity excited by the mere Contact of conducting Substances of different kinds.

In a Letter from Mr. Alexander Volta, F.R.S., Professor of Natural Philosophy in the University of Pavia, to the Rt. Hon. Sir Joseph Banks, Bart. K.B., P.R.S.

Read June 26, 1800.

Como in the Milanese, March 20, 1800.

After a long silence, for which I shall offer no excuse, I have the pleasure of communicating to you, Sir, and through you to the Royal Society, some striking results at which I have arrived in pursuing my experiments on the electricity excited by the mere mutual contact of metals of different kinds, and even by that of other conductors, also differing among themselves, whether liquids or containing some moisture to which they properly owe their conducting power.

The principal of these results, which encompasses nearly all the others, is the construction of an apparatus that resembles in its effects—that is to say, in the shocks it can impart to the arms, etc.—the Leyden jar, and even better, weakly charged electrical batteries, which would nevertheless act unceasingly, and whose charge after each explosion restores itself; which enjoys, in a word, an inexhaustible charge, a perpetual action or impulse on the electric fluid. But it differs essentially from them both by this continual action peculiar to it, and because, instead of consisting like ordinary Leyden jars and batteries of one or more insulating plates in thin layers (substances supposed to be the only electrics) armed with conductors, this new apparatus is formed solely of several of these latter bodies, chosen even from among the best conductors, and therefore the most remote, according to what has always been believed, from an electrical nature.

Yes, the apparatus of which I speak, and which will doubtless astonish you, is nothing but an assembly of a number of good conductors of different kinds arranged in a certain manner. Thirty, forty, sixty pieces or more of copper, or better of silver, each applied to a piece of tin, or what is much better, zinc, and an equal number of layers of water or some other liquid that is a better conductor than plain water, such as salt water, lye (*lessive*), etc., or pieces of cardboard, skin, etc., well soaked in these liquids; such layers being interposed between each couple or combination of the two different metals—such an alternating series, and always in the same order, of these three kinds of conductors: that is all that constitutes my new instrument.

This instrument imitates the effects of Leyden jars or electrical batteries by giving the same shocks; it falls, in truth, far below the activity of said batteries charged to a high degree as to the force and noise of the explosions, spark, and distance at which the discharge can occur, equaling only the effects of a battery charged to a very feeble degree, though having an immense capacity; but it infinitely surpasses the power of these batteries in that it does not need to be charged in advance by means of extraneous electricity, and is capable of giving shocks whenever it is touched properly, however frequent those contacts may be.

This apparatus—similar fundamentally, as I shall show, and even in form, to the natural electrical organ of the torpedo, electric eel, etc., far more than to the Leyden jar—I would like to call the **Artificial Electric Organ**. And is it not composed solely of conducting bodies? Is it not active of itself, without any previous charge, without the aid of any electricity excited by known methods, acting unceasingly, and capable at every moment of giving shocks, which, when frequently repeated or continued, produce that same numbness of the limbs that the torpedo causes?

Construction of the Vertical Column (The Pile)

I provide myself with several dozen small round plates or discs of copper, brass, or better silver, about an inch in diameter (such as coins), and an equal number of plates of tin, or much better zinc, of roughly the same shape and size. Exact precision is not required. I prepare, moreover, a sufficient number of discs of cardboard, leather, or other spongy matter capable of absorbing and

retaining much water or liquid. These wet discs I make a little smaller than the metal discs, so that when interposed, they do not project beyond them.

Having all these pieces ready—the metal discs clean and dry, and the non-metallic ones soaked in plain water or, much better, salt water, and lightly wiped so they do not drip—I place horizontally on a table one of the metal plates, for example silver, and on this first I adapt a second of zinc; upon this second I lay a wet disc; then another silver plate, followed immediately by another of zinc, succeeded again by a wet disc. I continue in this same manner, coupling a silver plate with one of zinc, always in the same order (silver below, zinc above, or vice versa), interposing a wet disc between each pair, building a column as high as can stand without falling.

If it contains about 20 of these pairs, it will already be capable of affecting Cavallo's electrometer (aided by the condenser) beyond 10 or 15 degrees, charging this condenser by a single touch so as to give a spark, and striking the fingers touching its two ends (the head and foot of the column) with small shocks.

To obtain these shocks, the fingers must be moistened with water so the skin is well soaked. To receive stronger shocks, one connects the bottom plate of the column with a basin of water in which one plunges one hand, while touching the top plate with a metal blade held firmly in the other wet hand. Touching successively the 3rd, 4th, 6th, and up to the 20th pair, the shocks increase gradually in strength, extending through the whole finger, the wrist, and even up to the elbow.

The Role of Electrolytes and Surface Area

The exact and extended contact of the wet discs is very important. While the metal plates need only touch at a few points (being excellent conductors), liquids and moist bodies are far less perfect conductors, requiring ample surface contact with the metals and one another so the electric fluid may pass without excessive resistance, especially when moved with such weak force.

The effects are considerably more sensible as the temperature is warmer, since heat renders water more conductive. But what makes it much better still are salts, particularly common salt.

If to the 20 pairs one adds 20 or 30 more, the shocks will extend through both arms to the shoulders, leaving the arm numbed if contacts are repeated rapidly.

The "Crown of Cups" (*Couronne de Tasses*)

I will now describe another form, which differs most in shape from the column, but which has the disadvantage of being more voluminous: the apparatus I call the **Crown of Cups** (Plate XVII, Fig. 1).

We arrange a row of several cups or goblets of any non-metallic matter (small drinking glasses are most convenient), half filled with salt water or lye. We link them all in a chain by means of metallic arcs, one arm *Aa* (immersed in one glass) being copper, brass, or silver-plated copper, and the other *Z* (in the next glass) being tin, or better zinc.

(I observe here that lye and alkaline liquors are preferable when tin is used; salt water is preferable when zinc is used).

The two metals of each arc are soldered together above the liquid. The submerged parts should have an area of about one square inch; the rest of the arc may be as narrow as desired, even a

simple wire.

A series of 30, 40, or 60 of these glasses forms this new apparatus, which in substance is identical to the column. Shocks are felt by plunging one hand into the first vessel and the other into the last.

Volta notes that if 60 cups are arranged in three groups of 20 with the middle 20 reversed in polarity, the shocks weaken and cancel out around the 40th cup before rising again at the 60th, demonstrating that the electromotive forces add algebraically.

Multi-Column Piles and Durability (Plate XVII, Figs. 2, 3, 4)

To make tall columns stable, Volta describes enclosing them within 3 or 4 vertical guide rods (*mmmm*) of glass or insulated wood (Fig. 2), or dividing the battery into multiple connected sub-columns (Figs. 3 & 4) linked by metal strips (*cc*).

To prevent the wet discs from drying out after a day or two, he recommends coating or potting the entire assembled cylinder in wax or pitch (*enveloppant de cire ou de poix toute la colonne*).

Sensory Effects on the Human Body

Volta details how continuous direct current affects the senses:

- **Touch & Continuous Pain:** When a wire from a 40–50 pair pile is pressed to moist skin (forehead, eyelid, or nose tip), there is an initial prickling shock upon making contact. If the circuit remains closed without interruption, this is followed by a persistent, sharp, burning pain that increases until it becomes unbearable. This proves the continuous, sustained circulation of the electric fluid. The pain is even more acute on broken skin, burns, or small cuts.
- **Taste:** Testing with zinc and silver on the tongue produces a distinct **acidic/sour taste** when the current enters the tip of the tongue (negative polarity), and an **alkaline/soapy/acrid taste** when the current leaves it. With 30–40 pairs, an immediate muscular convulsion and sharp prickle accompany the taste.
- **Sight:** Connecting the circuit across the moist face, eyelids, or mouth produces an instantaneous flash of light or luminous circle in the eyes upon making/breaking the loop.
- **Hearing:** Inserting two blunt metal rods deep into the ears and completing a 30-to-40-pair circuit produced a shock in the head, followed by a continuous crackling, bubbling, or boiling noise (like a thick paste boiling) that persisted as long as the circuit remained closed. Volta noted this sensation was disagreeable and feared it was dangerous, so he did not repeat it often.
- **Smell:** Attempting to stimulate the nose produced only localized stinging rather than any scent, because the electric fluid was contained in liquid conductors rather than discharging as a spark/effluvium through the air.

Conclusion: Refuting "Animal Electricity"

Volta concludes by examining the anatomy of the torpedo fish. He argues that William Nicholson's theory—that the fish's organ consists of tiny insulating talc plates acting as electrophores—is

impossible because living tissue is moist and conductive, incapable of holding electrostatic charges.

The torpedo's organ operates under the exact same principle as his own column: a large stack of conductive disks of differing materials whose mutual contact generates and sustains an electric current.

User prompt: He mentions using "Cavallo's electrometer" with a condenser, can you explain that?

Response:

- **Tiberius Cavallo's Electrometer:** A 18th-century sensitive device utilizing two small pith balls hung on fine linen threads or silver wires inside a glass bottle to detect small static charges. When charged, the pith balls repel each other, with the angle of separation indicating the voltage intensity.
 - **The Condenser (Capacitor) Attachment:** Because early chemical cells produced very low voltage (~1.1 V per cell) that was too weak to move pith balls directly, Volta attached a "condenser"—two insulated parallel brass discs separated by a thin layer of insulating varnish or resin.
 - **Accumulating the Charge:** By connecting one terminal of the pile to the lower disc and grounding the upper disc for a few moments, the condenser absorbed and stored the continuous current, accumulating minute amounts of charge over time.
 - **Reading the Result:** When the user quickly lifted the upper disc by its insulating handle, the capacitance dropped drastically, instantly multiplying the trapped voltage to a high enough potential to force Cavallo's pith balls to diverge past 10 or 15 degrees, making the weak current visible.
-

User prompt: Cavallo's paper mentions an electro phosphorous?

Response: **An electrophorus** (often spelled "electrophorus" or historically referred to with variants like *electrophore*) is a simple manual electrostatic generator invented in the late 18th century. It consists of an insulating resin plate and a metal plate with an insulating handle.

When the resin plate is rubbed to charge it via friction, the metal plate is placed on top, briefly touched to ground to drain induced charge, and then lifted away by its handle. The metal plate carries away a strong static charge via electrostatic induction without depleting the original charge on the resin, allowing the process to be repeated indefinitely.

In Volta's 1800 paper, he uses "electrophorus" and "condenser" concepts interchangeably in discussions regarding how static instruments function, and he heavily critiques contemporary rival theories (such as William Nicholson's) that attempted to model biological electric organs or chemical piles as networks of tiny built-in electrophores.

User prompt: What material was used to make the resin plate?

Response: The insulating plate (often referred to as the "cake") of an electrophorus was typically formed using a mixture of **shellac and resin**, or a specialized blend of **resinous materials, wax, and turpentine** spread over a base plate.

Alessandro Volta and other 18th-century experimenters favored these specific resinous compositions because they possessed exceptional dielectric properties, held a strong triboelectric negative charge when whipped or rubbed with fur, and retained that electrostatic charge for long periods without leaking away.

User prompt: I don't understand why the metal is connected to ground after being placed on the charged resin? Wouldn't that drain all the charge to ground?

Response:

- **Electrostatic Induction:** The charged resin plate creates an electric field that polarizes the metal plate placed on top of it, driving free electrons to the top surface while leaving a positive charge on the bottom surface facing the resin.
 - **The Grounding Step:** Connecting the metal plate to ground momentarily provides a path for those excess electrons on the top surface to escape into the earth, but the positive charge on the bottom remains trapped because it is held in place by the electrostatic attraction of the negative resin.
 - **Trapped Charge Retention:** The resin's own bound charge does *not* drain to ground because the resin is an insulator; the ground connection only interacts with the free electrons on the metal sheet.
 - **Lifting the Plate:** Once you break the ground connection and lift the metal plate away by its insulating handle, the net positive charge is trapped on the metal, giving you a high-voltage charge to use elsewhere while leaving the original resin cake completely untouched and ready to repeat the process.
-

User prompt: And this charge could be transferred to a capacitor in the form of a Leyden jar or something similar?

Response:

- **Direct Transfer:** Yes, lifting the charged metal plate and touching its edge to the knob of a Leyden jar transfers a small packet of high-voltage static electricity.
 - **Cumulative Charging:** Because the underlying resin cake does not lose its charge when the metal plate is lifted, you can repeat the sequence—place the plate, touch ground, lift, and touch the Leyden jar—dozens of times.
 - **Building a Charge:** Each cycle dumps another increment of charge into the Leyden jar, allowing you to accumulate a large static charge from a single initial rubbing of the resin cake.
-

User prompt: Earlier you mentioned condensers made of brass plates separated by varnish. I couldn't find any pictures like that?

Response: Historic images or search results for these specific devices are usually labeled under **"Volta's capacitor"** (or *condensatore elettrico*) and **"Volta's condensing electrometer"** (*elettroscopio condensatore*) rather than modern electronic capacitors.

In museum collections—such as the Tempio Voltiano in Como, Italy—these apparatuses typically look like:

- Two polished circular brass plates.
 - The lower plate sits fixed on an insulating glass or resin-coated pedestal.
 - The upper plate (the "shield") features an insulating glass or varnished handle extending upward.
 - Instead of modern plastic or ceramic dielectrics, the inner faces of the brass plates were lightly coated with an insulating layer of **shellac varnish, resin, or sealing wax**. This thin organic film prevented the bare metal faces from making direct galvanic or metallic contact when pressed together, maintaining a microscopic air and resin gap so static charges could "condense" via electrostatic induction.
-

User prompt: What recipe would Volta have used for the resin disks in his perpetual electrophore?

Response: Volta and 18th-century instrument makers did not use a single rigid chemical formula for the electrophorus resin cake, instead relying on combinations of natural resins, drying oils, and waxes melted together and poured into a shallow wooden or metal base.

- **The Core Ingredients:** The mixture typically combined **shellac** (for hardness and triboelectric charge retention) with **pine rosin** (colophony) and a small amount of **Venice turpentine** or **beeswax** to prevent the cake from becoming too brittle and cracking as it cooled.
- **Proportions:** A standard historical compounding ratio hovered around **2 parts shellac to 1 part rosin**, with just a few drops to a tablespoon of turpentine or wax added to plasticize the mixture.
- **Preparation Method:** The ingredients were melted together slowly in an iron crucible over low heat until completely liquefied, ensuring no water was trapped inside. The molten resin was then poured directly into a shallow circular wooden or metal pan (the *scutella*) to a depth of roughly 1/4 to 1/2 inch, where it solidified into a smooth, dark, highly insulating cake.

[Come costruire un elettroforo di Volta](#)

This video demonstrates a simple, modern replica of Volta's electrophorus using easily accessible household materials to illustrate the principles of electrostatic induction.

http://googleusercontent.com/youtube_content/1 *YouTube video views will be stored in your YouTube History, and your data will be stored and used by YouTube according to its [Terms of Service](#)*

User prompt: What is Venice turpentine and how was it made in the 18th and 19th centuries?

Response: **Venice turpentine** (also known as Venetian turpentine or larch balsam) is a thick, transparent, golden-yellow oleoresin. Despite its misleading name, it is **not** a distilled liquid solvent like modern gum turpentine; rather, it is the raw, viscous, honey-like sap of the European larch tree (*Larix decidua*). It earned its name because Venice served as the primary historic trading hub and gateway through which the resin was distributed across Europe.

Historical Harvesting Methods (18th–19th Centuries)

During the 18th and 19th centuries, harvesting larch resin was a specialized trade concentrated heavily in the alpine forests of Austria, the Tyrol, and northern Italy.

- **The Drilling Technique:** Unlike pine resin harvesting—which often involved stripping large sections of bark—larch harvesting relied on a specialized drilling method to tap the heartwood. Harvesters drilled an upward-slanting hole deep into the lower trunk of mature larch trees (usually between 1,000 and 1,300 meters above sea level) to reach the resin-bearing canals.
- **Collection Tubes:** A small wooden plug or hollow tube (often reed or elderberry pith) was inserted into the drilled hole to channel the slow-flowing, viscous balsam directly out of the tree.
- **Harvesting and Sealing:** Harvesters (historically referred to in regional traditions as drop hunters or woodsmen) periodically returned with small wooden casks or copper vessels to collect the accumulated resin. Once harvest season ended, the hole was plugged with a wooden peg or sealed with pitch to let the tree heal.
- **Refining:** The raw exudate was strained through coarse linen cloth to remove bits of bark, dirt, and wood dust, yielding a pure, honey-like balsam that retained its natural essential volatile oils without requiring artificial distillation.

[Larch Venice Turpentine - A Resin Not A Solvent](#)

This short video provides a quick visual overview of what Venice turpentine looks like and clarifies its true nature as a thick, viscous tree resin rather than a liquid solvent.

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User prompt: And was the shellac the same material as modern shellac used as a wood finish?

Response: **Yes, it is fundamentally the exact same material.**

Modern shellac flakes used by woodworkers today are still harvested from the exact same source: the secretion of the tiny lac bug (*Kerria lacca*) gathered from forest trees in India and Southeast Asia.

The primary differences between what Volta used for his electrophorus cakes and what you buy in a woodworking shop today come down to **processing and additives**:

- **Natural Wax Content:** Raw, unprocessed shellac naturally contains about 3% to 5% natural insect wax. For an 18th-century electrophorus cake or sealing wax, this natural wax was actually welcomed because it added flexibility and moisture resistance. For modern wood finishes, manufacturers often put the flakes through a chemical process to yield **dewaxed shellac**, which provides better clarity and allows subsequent topcoats (like varnish or polyurethane) to adhere properly.
 - **Color and Bleaching:** Historically, shellac came in deep ambers, browns, and oranges (commonly called "orange shellac") because of the twigs, tree sap, and natural pigments embedded in the raw resin. Modern woodworkers often use "blonde," "super blonde," or platinum grades that have been chemically bleached or filtered through activated carbon to remove those color bodies.
 - **Intended Use vs. Formulation:** While woodworkers dissolve shellac flakes in denatured alcohol to brush, wipe, or French-polish onto timber, Volta melted his flakes directly with rosin and turpentine into a thick, rigid structural mass meant to be poured solid into a mold.
-

User prompt: Is there any reason to prefer pine rosin that is already powdered over flakes or chunks in this application?

Response:

- **Faster Melting:** Powdered rosin has an immensely higher surface-area-to-volume ratio than large chunks or blocks, allowing it to liquefy almost instantly when heated in the crucible alongside shellac and Venice turpentine.
- **Easier Proportioning:** Fine powder can be measured by volume or weighed out precisely without needing a hammer or chisel to break down stubborn, glassy chunks of raw pitch.
- **Dust and Explosion Hazard:** Powdered rosin carries a severe airborne dust hazard; in a hot workshop or near an open flame used to melt the mixture, fine pine rosin dust is highly combustible and can flash or ignite explosively.
- **Oxidation:** Powdered rosin has vastly more surface area exposed to ambient air, making it more prone to oxidation and moisture absorption over time compared to dense, solid blocks or flakes.

Recommendation: Chunks or coarse flakes are generally preferred for this application. They melt down smoothly with minimal stirring while completely avoiding the explosive dust and rapid oxidation risks associated with fine powder.

User prompt: I found some "larch gum" that says it is made from the resin of larch trees. Larch turpentine returns results of distilled gum turpentine and Venice turpentine the same or a horse remedy for coating hooves

Response: **Larch gum** (often sold under names like American larch gum or larch arabinogalactan) is chemically different from **Venice turpentine** and gum turpentine.

- **Larch Gum (Arabinogalactan):** This is a water-soluble polysaccharide extracted from the wood of western larch trees (*Larix occidentalis*). It dissolves completely in water and acts as a binder, dietary fiber, or emulsifier, but it has no resinous, sticky, or waterproofing properties when dry.
- **Venice Turpentine (Oleoresin):** As discussed earlier, this is the raw, sticky oleoresin tapped directly from the heartwood of the European larch tree (*Larix decidua*). It is hydrophobic, viscous, and soluble in solvents like spirits of turpentine or alcohol, making it suitable for hardening resin blends.
- **Distilled Gum Turpentine:** This is the volatile solvent fraction produced by steam-distilling raw pine or larch tree resins, yielding a thin liquid spirit used for thinning paints and varnishes.
- **The Hoof Remedy Connection:** The horse hoof remedy reference comes from historical veterinary practices where thick larch or pine turpentines were painted onto equine hooves to provide a flexible, waterproof barrier against moisture and thrush.

What you need for an electrophorus cake: You need an oleoresin or balsam (like genuine Venice turpentine) rather than water-soluble larch gum or distilled liquid gum turpentine.

User prompt: So the horse hoof coating is the closest? Could i use beeswax to plasticize the cake instead? I am not sure about the nature of these other products though at least one is described as an oleoresin

Response: **Yes, beeswax is historically correct and works well as a plasticizer.**

While Venice turpentine adds a sticky, tacky flexibility, 18th-century makers and modern historical craftsmen frequently used pure beeswax or tallow to soften brittle resin-and-shellac blends and prevent them from cracking under mechanical stress.

- **The Equine Connection:** The reason the horse hoof coating matches chemically is because historical veterinary "Venice turpentine" was authentic larch oleoresin; because horseshoes and hooves needed a flexible, waterproof dressing that wouldn't wash off in mud, the exact same raw balsam harvested from the Alps was sold for both artistic/pharmaceutical and veterinary uses.
 - **Beeswax vs. Turpentine:** Beeswax will make the cake slightly softer and less sticky than turpentine, so you only want to use a small proportion (roughly 5% to 10% by weight of the total shellac/rosin mix) to avoid making the surface too soft to hold a static charge.
-

User prompt: I noticed that Cavallo's electrometer and Volta's capacitor electrometer both have sealing wax on the top neck of the glass bottle. What's the value of that?

Response: **Sealing wax on the neck of an electrometer bottle acts as an absolute moisture barrier and mechanical anchor.**

- **Preventing Surface Leakage:** Glass readily attracts an invisible microscopic film of moisture from ambient air. This moisture creates a conductive path that allows static charges to bleed away into the air or down the outside of the bottle, destroying the instrument's sensitivity.
 - **Hermetic Insulation:** Sealing wax is a hydrophobic, high-dielectric resin blend. Melting it around the neck where the metal support rod or cork enters the glass hermetically seals the joint against humidity.
 - **Mechanical Stability:** It rigidly locks the internal suspension wire, brass rod, or hook in place, preventing it from shifting or rotating when the instrument is handled or connected to a charging lead.
-

User prompt: That's ... interesting. Some versions of electrometer used gold leaf. Could that be replaced with aluminum foil?

Response: Aluminum foil works mechanically, but its high mass creates severe operational limitations compared to traditional gold leaf.

- **Weight and Thickness:** Standard household aluminum foil is roughly 0.016 mm (16 microns) thick and weighs about 43 g/m^2 . Historical gold leaf used in electrometers was hammered down to roughly 0.0001 mm (0.1 microns)—over 100 times thinner—and weighed a fraction of a gram per square meter.
- **Inertial Resistance:** Because aluminum foil is vastly heavier and stiffer, it requires a much stronger electrostatic charge to overcome its own mass and push the leaves apart, making the electrometer significantly less sensitive to weak static fields.
- **Tearing and Creasing:** Gold leaf behaves almost like a fluid sheet, fluttering smoothly under minute electrical attractions. Aluminum foil tends to crease, kink, and hold rigid folds that interfere with smooth divergence.

Practical Workaround: If you want to build a working replica without tracking down genuine gold leaf, look for **ultra-thin aluminum condenser microphone diaphragms**, extremely thin metallic gift-wrapping foil, or carefully split multi-layer foil wrappers, keeping the hanging strips as short and narrow as possible to minimize weight.

User prompt: What about craft gold leaf sheets? Are they thin enough?

Response: **Yes, craft gold leaf sheets can work**, but you need to check whether you are buying **genuine gold** craft leaf or **imitation** craft leaf (often called composition or Dutch metal), as their physical properties differ significantly for an electrometer:

- **Genuine Craft Gold Leaf:** If the craft book contains real gold (even lower karats like 12k or 22k), it is beaten down to roughly 0.1 to 0.5 microns thick. This is practically identical to historical leaf and will perform wonderfully with extreme sensitivity.
- **Imitation Craft Leaf (Composition / Dutch Metal):** Most cheap craft store "gold" or "silver" leaf is actually an alloy of copper and zinc (for gold) or thin aluminum (for silver). While

genuine imitation gold is still quite thin (roughly 0.8 to 1.0 micron), it is noticeably stiffer, heavier, and springs back rather than fluttering fluidly.

If you use imitation craft leaf, the electrometer will still work, but you will need a stronger electrostatic charge to overcome the stiffness and mass of the strips compared to true beaten gold leaf.

User prompt: So what is the difference between 24 carat and 12 carat?

Response:

- **Purity:** 24-carat gold is 100% pure gold (or 24 parts out of 24), whereas 12-carat gold is only 50% pure gold (12 parts gold mixed with 12 parts of other metals like copper, silver, or zinc).
 - **Properties:** Because pure gold is extremely soft and malleable, 24-carat gold bends and scratches easily. Mixing it down to 12-carat adds alloy metals that drastically increase structural strength, durability, and scratch resistance.
 - **Color and Workability:** Lower-carat golds can vary in color (such as yellow, white, or rose) depending on the specific alloy metals used, and they handle differently when beaten into ultra-thin leaf or worked by a jeweler.
-

User prompt: Why would they put foil on the inside of the bottle?

Response:

- **Capacitance and Conduction:** A Leyden jar requires both an inner and outer conductive layer to act as the "armatures" (plates) of a capacitor, allowing opposite electrical charges to accumulate on either side of the glass.
 - **Charge Storage:** The internal metal foil lining serves as the internal conductor that receives and stores the charge introduced through the top chain or rod.
 - **Glass as a Dielectric:** The glass bottle itself acts as the insulating dielectric barrier, keeping the charges separated until a discharge occurs.
-

User prompt: Oh I meant inside an electrometer.

Response:

- **Eliminating Internal Static Drag:** The inside of an electrometer bottle is lined with strips or patches of foil (or coated in conductive tinfoil/graphite) to create an equipotential Faraday shield or to ground the interior glass walls.
- **Preventing Charge Sticking:** Uncoated glass inside an electrometer can trap stray static charges on its surface from induction, which then attract, pin down, or deflect the delicate gold

leaves, causing erratic or stuck readings.

- **Controlled Grounding:** In some designs (like Kelvin or quadrant electrometers), the interior lining is connected to ground, ensuring that the leaves interact only with the central charge terminal rather than unpredictable static fields clinging to the inner glass walls.
-

User prompt: I'm curious because that's the only element that needs the bottom of the bottle removed.

Response:

- **Grounding the Enclosure:** Grounding strips or inner foil linings on an electrometer (such as a Henley quadrant electrometer or specialized Henley-style designs) create a conductive shield that protects the interior environment from erratic external electrostatic fields.
 - **The Bottom-Removal Need:** Constructing certain historical electrometers required removing the bottom of a glass bottle or bell jar to physically install internal mounting fixtures, support pillars, or graduated scale arcs that could not fit through the narrow neck.
 - **Sealing It Back Up:** Once the internal components, plumb lines, or indicator needles were secured to the base or a bottom wooden/metal plate, the jar was sealed back up to maintain an airtight, dust-free chamber protecting the delicate moving parts.
-

User prompt: What is type III glass? I can find some cheap round bottles.

Response:

- **Soda-Lime-Silica Glass:** Type III glass is standard commercial soda-lime glass, which is the most common and inexpensive glass used for everyday bottles, jars, and mass-produced beverage containers.
 - **Chemical Resistance:** It has moderate hydrolytic resistance, meaning it is fine for general liquids, foods, and many chemical solutions, but it can leach minor amounts of alkali into highly sensitive or acidic reagents compared to lab-grade borosilicate (Type I).
 - **Thermal Performance:** It has a high thermal expansion coefficient, making it vulnerable to thermal shock (cracking if heated or cooled too rapidly).
 - **Affordability:** Because it is mass-produced for commercial packaging, round Type III glass bottles are widely available and extremely inexpensive, making them a practical choice for bulk storage or DIY projects where extreme thermal shock or ultra-pure chemical inertness isn't required.
-

User prompt: How easy is it to cut off the bottom of a glass bottle?

Response: Cutting the bottom off a standard Type III soda-lime glass bottle is moderately difficult and requires patience, but it is entirely doable in a home workshop using a few standard methods.

Because soda-lime glass is thermally sensitive, it behaves very differently than specialized borosilicate glass.

- **Thermal Shock Scoring (The String Method):** Wrap a flammable string (soaked in alcohol or acetone) tightly around the bottle where you want the cut. Ignite the string, let the flame heat the glass evenly for a minute, and then immediately plunge the bottle into a bucket of cold water. The sudden thermal contraction often causes a clean crack to propagate along the heated line, though the edge will be sharp and rough.
- **Scoring and Temperature Cycling:** Score a continuous scratch around the circumference of the bottle using a glass-cutting wheel or a diamond scribe. Alternately apply heat (using a localized torch or boiling water) and a cold source (like an ice cube or cold damp sponge) directly over the score line until the bottom thermal-fractures off.
- **Diamond Grit Tile Saw or Wet Saw:** For the cleanest, straightest cut, use a wet tile saw equipped with a continuous-rim diamond blade. Support the bottle carefully, feed it slowly into the spinning blade with a steady stream of water cooling the cut, and let the diamond grit grind through the soda-lime glass without forcing it.

Regardless of the method you choose, always wear eye protection and heavy gloves, and expect to clean up the newly exposed edge with wet-dry silicon carbide sandpaper (around 80-grit to 220-grit) to remove sharp burrs.

User prompt: These would all be possible with square bottles too.

Response:

- **Flat Surfaces:** Square bottles provide flat exterior walls, making it significantly easier to adhere conductive tinfoil or copper tape evenly to both the inside and outside when building a square Leyden jar or capacitor.
 - **Mechanical Stability:** A square or rectangular cross-section prevents the vessel from rolling around on the workbench during assembly, wiring, or mounting operations.
 - **Internal Alignment:** Flat inner walls allow internal foil linings or electrode plates to sit flush against the glass without awkward curving air gaps.
 - **Consistent Scoring:** Using a glass-scoring tool or thermal wire on square glass requires careful handling around the sharp 90-degree corners, as thermal cracks tend to wander unpredictably away from rounded radii unless scored deeply.
-

User prompt: I was thinking about a wet tile saw

Response:

- **Clean, Flat Cuts:** A wet tile saw with a continuous-rim diamond blade yields a perfectly straight, square edge on glass bottles without the wandering thermal fractures common with the string method.

- **Corner Navigation on Square Bottles:** Feeding a square bottle through a wet saw requires you to rotate the bottle slowly through the blade at each 90-degree corner, letting the diamond grit grind the flat face before navigating the radius.
 - **Thermal Stress Management:** The constant stream of cooling water on a wet tile saw drastically minimizes local thermal stress, preventing the soda-lime glass from cracking unpredictably during the cut.
 - **Edge Cleanup:** Even with a wet saw, the cut glass will leave a micro-serrated edge; always back up the cut with wet-dry silicon carbide sandpaper (around 120-grit to 220-grit) lubricated with water to ease the sharp rim.
-

User prompt: Ok I am thinking about the order in which I should reconstruct these things. Seems like the electrophore is a good starting point, followed by a Leyden jar or electrometer, and a capacitor.

Response: **1. Volta's Electrophorus (The Generator)** Starting here is essential because it provides your primary, renewable source of electrostatic charge via induction. Without a reliable way to generate static electricity, you cannot test or charge any subsequent apparatus.

2. Gold-Leaf Electrometer (The Detector) Building this second gives you an immediate tool to visualize and measure the charges coming off your electrophorus. You need a sensitive detector to confirm that your resin cake is charged, your metal plate is working, and your insulation is holding.

3. Leyden Jar (The Storage Tank) Tackling the Leyden jar third lets you capture and accumulate the incremental charge packets generated by your electrophorus. It turns a momentary static spark into a heavy, high-voltage jolt you can store and discharge at will.

4. Volta's Condenser / Brass-Plate Capacitor (The Fine Instrument) Saving the multi-plate brass condenser for last makes the most sense because it combines the principles of all the previous devices. It relies on the induction concepts of the electrophorus, the storage principles of the Leyden jar, and the detection sensitivity of the electrometer to demonstrate how minute charges are "condensed" into measurable potentials.

User prompt: And then the voltaic pile, crown of cups, and trough battery. Or was there any other experiments in between voltas paper and the trough battery?

Response:

- **Simultaneous Reveal:** In Volta's famous March 1800 announcement letter to the Royal Society of London ("*On the Electricity Excited by the Mere Contact of Conducting Substances*"), he didn't just introduce the vertical **voltaic pile**—he debuted the **crown of cups** (*couronne de tasses*) right alongside it in the exact same paper. He viewed them as two configurations of the same underlying "apparatus of electromotive force."
- **The Intervening Gap:** There were no major revolutionary battery architectures invented *between* Volta's paper and the trough battery. Instead, the immediate aftermath of his 1800 publication triggered an explosion of replication, experimentation, and minor mechanical refinements across Europe.

- **The Trough Battery Evolution:** The **trough battery** (or "trough validator") was actually created shortly after by British chemist **William Cruickshank** in 1800/1801. Cruickshank didn't change the chemistry; he solved a frustrating mechanical flaw of Volta's vertical pile: the extreme weight of tall stacks squeezed the brine out of the cardboard spacers, causing them to dry out, leak conductive salts down the sides, and short-circuit. By cementing the alternating copper and zinc plates vertically into slots inside a long, horizontal wooden trough (creating water-tight compartments filled with acid or brine), he made a much more stable and robust battery.

Sources

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 2. [Volta Invents the Battery | History | Research Starters - EBSCO](#)
 3. [History of the battery - Wikipedia](#)
-

User prompt: And the Daniell cell was the next big breakthrough?

Response: **Solving Polarization** The Daniell cell, invented by British chemist John Frederic Daniell in 1836, was indeed the next major technological leap because it permanently solved the crippling issue of **polarization**.

The Hydrogen Problem Volta's pile, the crown of cups, and Cruickshank's trough all suffered from a shared operational flaw: during discharge, hydrogen gas bubbles accumulated and clung to the copper plate. This insulating blanket of gas rapidly increased internal resistance and caused the battery's voltage to plummet within minutes under a heavy load.

The Two-Fluid Breakthrough Daniell eliminated hydrogen gas entirely by splitting the cell into a **two-fluid system** separated by a porous barrier (such as unglazed earthenware or an ox-gullet membrane):

- **The Anode Side:** A zinc rod immersed in a zinc sulfate or dilute sulfuric acid solution.
- **The Cathode Side:** A copper plate or container immersed in a copper sulfate solution.
- **The Reaction:** Instead of generating escaping gas, copper ions from the solution safely plated themselves directly onto the copper electrode as pure metal, keeping the cell's electrical output remarkably stable.

Impact on Infrastructure Because it was the first battery capable of providing a steady, continuous current over hours and days rather than minutes, the Daniell cell became the absolute workhorse of the 19th-century electrical revolution. It powered the expansion of global telegraph networks and was even used as the voltage standard to help define the modern volt.

Sources

1. [The Daniell Cell: A More Reliable Source of Electricity](#)
 2. [Daniell Cell - Magnet Academy - National MagLab](#)
 3. [Telegraph Batteries: Daniell and Grove Cells Explained - FlashFish](#)
-

User prompt: Ox gullet! Is that the same as the cow tracheas available as dog treats?

Response: **No, an ox gullet is the esophagus, whereas a cow trachea is the windpipe.**

While both are tubular animal tissues commonly sold today as dehydrated dog chews, they differ in both biological structure and historical function:

- **Ox Gullet (Esophagus):** The esophagus is a muscular, flexible tube lined with a smooth mucosal membrane. Historically, this muscular and fibrous tissue was cleaned, split open, and dried to create thin, surprisingly tough, semi-permeable membranes. They were used for everything from sausage casings to early biological diaphragms and porous partitions in chemical cells.
- **Cow Trachea (Windpipe):** The trachea is reinforced by rigid, horseshoe-shaped rings of cartilage designed to keep the airway open against air pressure. While it is tubular, those stiff cartilage rings make it entirely unsuitable for flattening out or using as a flexible, uniform porous membrane in an electrochemical cell.

If you are looking to replicate an authentic porous barrier for a historical cell variant like an early Daniell or porous-cup cell, a thin slice of unglazed earthenware (like a flowerpot saucer) or a modern ceramic membrane will give you a much more predictable and stable barrier than organic membranes.

User prompt: How about advances in measuring current?

Response: The evolution of current measurement mirrors the shift from qualitative detection to precise, standardized engineering, moving alongside the battery breakthroughs of the 19th century:

- **The Oersted Effect & The "Multiplier" (1820):** Immediately after Hans Christian Ørsted discovered that a live wire deflects a compass needle, Johann Schweigger created the first crude current detector by wrapping a coil of wire many times around a compass. Because the turns multiplied the magnetic effect, early scientists called it a "multiplier".
- **The Tangent Galvanometer (1837):** Developed by Claude Pouillet, this was the first truly practical instrument for *quantifying* current. By aligning a vertical coil with Earth's magnetic field, the current passing through could be mathematically calculated based on the tangent of the deflection angle of the needle.
- **The Mirror Galvanometer (Late 1850s):** Invented by William Thomson (Lord Kelvin) to detect extremely faint signals on the first transatlantic telegraph cables, this replaced physical pointers with a tiny mirror and a beam of light. A microscopic twist of the suspended magnets threw a light-spot across a distant scale, magnifying sensitivity by orders of magnitude.
- **The d'Arsonval Moving-Coil Meter (1882):** Jacques-Arsène d'Arsonval revolutionized the field by flipping the design: instead of a moving magnet affected by Earth's field, he used a stationary permanent magnet and suspended a **moving coil of wire**. Soon commercialized by Edward Weston, this laid the direct structural foundation for nearly all analog multimeters, ammeters, and voltmeters used through the 20th century.

Sources

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 2. [A Short History of the Galvanometer](#)
 3. [Tangent Galvanometer - Washington and Lee University](#)
 4. [Galvanometer - ELECTRIC WORLD](#)
-

User prompt: What was the next battery breakthrough after the Daniel cell

Response:

- **The Grove Cell (1839):** Invented by Welsh physicist William Robert Grove, this two-fluid cell substituted the copper component of the Daniell cell with a platinum cathode immersed in concentrated nitric acid. The nitric acid acted as a powerful depolarizer, dramatically lowering internal resistance and boosting output voltage from 1.1 volts up to nearly 1.9 volts, making it the premier power source for early telegraph lines despite emitting noxious nitric oxide fumes.
- **The Bunsen Cell (1841):** German chemist Robert Bunsen made Grove's high-voltage design economically practical by replacing the expensive platinum cathode with cheap, molded carbon blocks sourced from coal-gas retorts. This maintained the powerful output of the Grove cell while slashing manufacturing costs.
- **The Lead-Acid Battery (1859):** French physicist Gaston Planté created the world's first rechargeable (storage) battery. By submerging two spiral-wound lead sheets separated by cloth into a bath of dilute sulfuric acid and running a charging current through them, he built a cell that could store and return large surges of electrical energy. This direct ancestor of modern automotive batteries solved the permanence problem of primary cells.
- **The Leclanché Cell (1866):** Developed by French engineer Georges Leclanché, this cell paired a zinc anode with a porous cup containing a manganese dioxide and carbon mixture, using an ammonium chloride solution as the electrolyte. It was far more stable, portable, and user-friendly than its predecessors. Its later refinement into the 19th-century "dry cell" enabled the mass commercialization of portable consumer electronics.

Sources

1. [Battery - Rechargeable, Storage, Power | Britannica](#)
-

User prompt: The le clanche cell wasn't rechargeable

Response: The Leclanché cell is indeed a primary, single-use battery rather than a rechargeable secondary cell.

While Planté's lead-acid battery solved the rechargeability problem a few years earlier, it was heavy, hazardous, and impractical for low-drain household devices. The Leclanché cell traded rechargeability for sheer practical convenience:

- **Stable Shelf Life:** Because its components remained chemically stable when the circuit was open, it didn't self-discharge rapidly like earlier wet cells.
- **Lower Maintenance:** It replaced aggressive acids with a milder ammonium chloride solution.

- **The Dry Cell Evolution:** Its architecture was later modified by Carl Gassner into the **zinc-carbon dry cell**, replacing the liquid solution with an immobilizing paste that prevented spills and enabled mass-market portable electronics.

Sources

1. [Leclanche battery - Simple English Wikipedia, the free encyclopedia](#)
 2. [leclanche cell meaning - English dictionary | VDict](#)
-

User prompt: What did Plante make the “protective box” of when he demonstrated his battery with 9 cells in parallel

Response: When Gaston Planté presented his famous 9-cell secondary battery arrangement to the French Academy of Sciences in 1860, the protective box housing the elements was made of **tarred pitch-pine** (*pitch-pin goudronné*).

The interior of this wooden box was divided to hold the glass or hard rubber vessels containing the spiral lead elements, and the wood was heavily treated with pitch (tar) to protect it from being degraded by sulfuric acid splashes and leaks.

Sources

1. [Philippe B. de l'Arc - Histoire de la voiture électrique - Philippe Boursin](#)
-

User prompt: I read that an improved version of the Plante cell coated the plates with lead oxide. Doesn't that form on the lead plate automatically when charging?

Response: Yes, lead dioxide forms automatically on the positive plate when charging, but the shortcut you are thinking of—manually coating the plates with lead oxide *before* assembly—was the breakthrough invention introduced by **Camille Alphonse Faure in 1881**, over two decades after Planté's original design.

- **Planté's Original Method ("Forming"):** Planté started with plain, smooth sheets of rolled lead. When you first charge a plain lead cell, only the microscopic outermost skin of the metal turns into lead dioxide (PbO_2), yielding a very low electrical capacity. To make it useful, Planté had to put the battery through a long, tedious "forming" process—repeatedly charging it, discharging it, reversing the current, and letting it rest over weeks or months. This slow electrochemical corrosion ate deeper into the lead, slowly building up a thin, deeply textured layer of active lead oxides and dramatically increasing the surface area.
- **Faure's Paste Method:** Faure realized he could completely bypass months of tedious electrochemical "forming" by physically pasting a layer of lead oxides (red lead and litharge mixed with sulfuric acid) directly onto a cast lead grid. When first charged, that pre-existing paste instantly converted into active sponge lead and lead dioxide, yielding a battery with massive capacity right out of the gate and making mass-production commercially viable.

Sources

1. [Lead-acid battery - Wikipedia](#)

2. How Lead Acid Battery Grids Are Made Step by Step

User prompt: I saw a diagram of a pair of needles mounted vertically so that the points point towards each other and a spark formed when connected to a battery.

Response: That configuration is a classic **needle spark gap** designed to demonstrate or utilize concentrated electrical discharge.

- **The Physics of Sharp Points:** Sharp points concentrate electric field lines intensely because of their tiny radius of curvature. This high field gradient drastically lowers the threshold voltage needed to ionize the surrounding air molecules, allowing a spark to bridge the gap much more easily than it would between flat or rounded terminals.
 - **The Voltaic Pile Context:** While a low-voltage battery cannot jump an air gap on its own, a large historical **voltaic pile** consisting of 50 to 100 or more series-connected plates generates enough cumulative electromotive force to spark across a tiny gap when the circuit is manipulated.
 - **Make-and-Break Inductive Kicks:** In many 19th-century illustrations, bringing pointed conductors connected to a battery close together (or snapping them apart) demonstrated the bright spark caused by self-induction—the sudden high-voltage spike created by the collapsing magnetic field in the circuit wires when the current is interrupted.
-

User prompt: Hmm, so if the needles are connected there is a current flowing from the battery. When the connection is broken the magnetic field created by the current collapses. That creates the high voltage for a spark?

Response: **The exact physical mechanism is self-induction, often referred to as an "inductive kick" or flyback voltage.**

- **Magnetic Energy Storage:** Any circuit containing wire loops or coils possesses inductance (L). When a closed circuit allows current (I) to flow from a battery, it builds up a surrounding magnetic field that stores energy ($\frac{1}{2}LI^2$).
- **The Rate of Current Change ($\frac{di}{dt}$):** When you abruptly snap the connection apart, the time it takes for the current to drop to zero is vanishingly small. This creates a massive, negative rate of change in current ($\frac{di}{dt}$).
- **Faraday's Law of Induction:** According to electromagnetic induction, the voltage generated across an inductor is proportional to the rate of change of current:

$$V = -L \frac{di}{dt}$$

Because $\frac{di}{dt}$ is so enormous during a sudden break, the induced voltage V spikes well above the original battery voltage—sometimes reaching hundreds or thousands of volts.

- **Overcoming Air Breakdown:** That sudden voltage spike forces electrons to bridge the tiny air gap between the sharp needle points, ionizing the air molecules and creating a brief, bright spark before the magnetic field completely collapses.
-

User prompt: How does it work going the other way? Have to close it quickly enough to make Di/dt large?

Response: When you close a circuit containing inductance, the electrical dynamics behave in the exact opposite way, governed by the growth of current rather than its collapse.

- **The Back-EMF Barrier:** When you push the needles together to close the circuit, current attempts to jump from zero to its maximum value. Because an inductor resists any change in current, it instantly generates a **back-EMF** (an induced voltage pointing in the opposite direction of the battery) to fight that increase:

$$V = -L \frac{di}{dt}$$

- **Delayed Current Rise:** While the initial mathematical rate of change ($\frac{di}{dt}$) is steep the moment contact is established, the resulting induced voltage acts *against* the battery source. Instead of creating a massive voltage spike, it chokes and smooths out the incoming current, forcing it to ramp up gradually according to the circuit's time constant ($\tau = \frac{L}{R}$).
- **Pre-Contact Arcing:** If you notice a tiny spark the instant you *close* a pair of needles, it is generally not an inductive kick. Instead, it is a microscopic electrostatic pre-discharge—the high electric field ionizes the air gap a split second before the metal actually touches.

The premise that closing a circuit quickly creates a large $\frac{di}{dt}$ inductive spike is **incorrect**. The massive, high-energy voltage spike and bright spark occur exclusively when *breaking* the circuit, because the collapsing magnetic field has nowhere else to force its energy except across the opening air gap.

User prompt: How did people set up the circuit to have a high inductance?

Response: Nineteenth-century experimenters maximized circuit inductance by combining multi-turn coil geometry, high-permeability cores, and specialized winding configurations:

- **Densely Wound Solenoids (Helixes):** Because inductance scales with the square of the number of turns (N^2), builders like Joseph Henry and Charles Grafton Page wound thousands of feet of insulated copper wire into tight, multi-layered cylindrical bobbins to concentrate the magnetic field.
- **Bundled Iron Cores:** Inserting a metal core dramatically increases inductance by concentrating magnetic flux, but a solid iron bar suffers from severe energy losses due to internal circulating currents (eddy currents). Experimenters solved this by stuffing the core with a **bundle of parallel, insulated soft-iron wires**, which preserved high magnetic permeability while blocking parasitic electrical currents.

- **Coaxial Primary and Secondary Layers:** Pioneers like Nicholas Callan and Heinrich Ruhmkorff nested coils together. They used a short, thick primary wire connected to the battery to generate a massive magnetic flux, surrounded by an insulated secondary winding made of miles of ultra-thin wire to maximize the inductive coupling and voltage transformation.

Sources

1. Induction Coil - Physics - Kenyon College

User prompt: The diagram I see just had loosely coiled or spiral wires. Would that be enough to create inductance?

Response: Any conductor carrying electric current possesses some degree of self-inductance, but changing its geometry into a loop or a spiral dramatically amplifies that effect.

- **Magnetic Field Reinforcement:** When a wire is straight, its magnetic field lines disperse widely into space. Bending that same wire into a loose coil or spiral forces the magnetic field lines from adjacent turns to overlap and reinforce one another, effectively concentrating the magnetic flux within and around the coil.
 - **The Turn-Count Scaling:** Inductance scales with the square of the number of turns (N^2) and is proportional to the area enclosed by the loop. Even a loosely wound spiral or a helix with just a few dozen turns stores substantially more magnetic energy than a straight wire of identical length.
 - **Historical Effectiveness:** Early 19th-century experimenters routinely used simple, hand-wound copper spirals or loose bobbins connected to a multi-cell voltaic battery. When the circuit was abruptly broken, the magnetic energy stored even in a modest, loose coil was more than sufficient to produce a sharp, bright inductive spark across a contact gap.
-

User prompt: Are the electrodes in a modern lead acid battery made of sponge lead and lead dioxide?

Response: The negative electrode of a modern lead-acid battery consists of porous metallic lead, commonly referred to as **sponge lead**, while the positive electrode is made of **lead dioxide** (PbO_2).

Both active materials are pasted onto cast lead-alloy grids to provide mechanical support and current collection, and they are submerged in an electrolyte solution of dilute sulfuric acid (H_2SO_4). During discharge, both electrodes react with the sulfuric acid to form lead sulfate (PbSO_4), and during charging, the reverse reaction converts them back into sponge lead and lead dioxide.

User prompt: The lead sulfate is in solution? And plates back into metal during charging?

Response: Lead sulfate is **not** dissolved in the liquid electrolyte. Because it has an extremely low solubility in sulfuric acid, it precipitates out as a solid, fine-grained white crystalline deposit that clings directly to and fills the porous interior of both electrode plates.

During the charging cycle, this solid layer does not behave like a standard metal-plating bath where dissolved ions travel through a liquid and plate onto a cathode. Instead, it undergoes a solid-state electrochemical conversion:

- **The Negative Plate:** Solid lead sulfate is reduced directly back into porous metallic sponge lead, releasing sulfate ions back into the surrounding liquid.
- **The Positive Plate:** Solid lead sulfate is simultaneously oxidized into solid lead dioxide, also returning sulfate ions to the electrolyte and restoring the strength of the sulfuric acid.

The premise that lead sulfate floats freely in solution and plates back into metal is **incorrect**. It remains a solid mineral deposit on the plates that chemically transforms back into active electrode material during charging.

User prompt: Ah, so that's why lead acid cells don't cope with deep discharge? What is different in a deep cycle battery?

Response:

- **Plate Thickness and Surface Area:** Starting (SLI) batteries use a high count of thin, highly porous plates to maximize surface area for the massive, instantaneous current required to crank an engine. Deep-cycle batteries swap this architecture for fewer, much thicker, heavier plates, trading away high immediate cranking power for structural endurance.
- **Volume Expansion and Mechanical Stress:** When lead and lead dioxide convert into lead sulfate during discharge, the material expands significantly in volume. In a starting battery, this heavy volume shift physically strains, warps, and shreds the delicate thin plates, causing the active material to flake off ("shed") and drop uselessly to the bottom of the case. Thick plates in deep-cycle batteries absorb this mechanical stress without fracturing.
- **Hard Sulfation Prevention:** Deep discharges leave behind massive amounts of lead sulfate. If allowed to sit on thin, fragile plates, those tiny crystals grow into large, stubborn, electrically insulating crystals ("hard sulfation") that refuse to convert back during charging. Deep-cycle batteries utilize denser active pastes, heavier grid alloys, and often glass-mat (AGM) or gel containment to inhibit this destructive crystal growth, allowing them to safely cycle down to 50% or 80% capacity repeatedly.

Sources

1. [Deep Cycle vs Starting vs Dual Purpose Batteries - Midtronics](#)
2. [\[Full Guide\] Starting Battery vs. Deep-Cycle Battery - LiTime](#)
3. [Deep Cycle vs Starting Battery: Which Is Right for You?](#)

User prompt: Were the spark gaps spring loaded or how did they break the gap quickly?

Response: Early 19th-century experimenters used a mix of clever manual techniques and early electromechanical devices to break circuits fast enough to generate sharp inductive sparks.

- **The File-and-Wire Method:** Michael Faraday famously used a simple metal file connected to one battery terminal while drawing a wire across its textured surface. The wire skipped rapidly over the teeth, creating hundreds of microscopic, high-speed breaks a second that produced a continuous, hissing stream of bright inductive sparks.
- **Spring-Loaded Automatic Interrupters (Neef's Hammer):** Invented in the 1830s by Christian Ernst Neef, this became the standard mechanism for continuous induction coils. A spring-steel armature holding a contact point was pulled toward the iron core by its magnetic field the moment the circuit closed. That physical movement broke the contact, killing the magnetic field, which caused a spring to instantly snap the contact back closed to restart the cycle.
- **Spring-Loaded Hand Keys:** For single, deliberate electrostatic or inductive "kicks," builders used spring-loaded brass lever switches and telegraph-style keys. The tensioned spring forced the contacts apart much faster than human fingers could pull them, ensuring a steep, clean $\frac{di}{dt}$ voltage spike.

Verification: You can verify a clean, high-speed break by watching for a sharp, crisp snap and a bright, localized point-source spark rather than a dull, lingering orange arc that drags out the transition time.

What kind of circuit break mechanism are you planning to incorporate into your historical build?

User prompt: In this circuit diagram there is a "condenser" - in this likely 2 brass discs separated by resin ? Or something else and what is the purpose in the induction circuit?

Response:

- **What it's actually made of:** Unlike Volta's electrostatic condensers (which sometimes used brass discs and resin), the condenser in a historical induction coil is an early parallel-plate capacitor constructed from alternating sheets of **tinfoil interleaved with paraffin-waxed paper, oiled silk, or mica**. They were often stacked flat inside the base of the coil box or rolled up tightly.
- **Preventing Contact Arcing:** When the automatic interrupter (Neef's hammer) snaps open, the massive inductive "kick" tries to maintain the current by forcing a spark to jump across the opening contact points. This lingering arc would slowly burn away the platinum points and bleed off energy. The condenser provides an immediate local storage reservoir, absorbing that initial surge so the contacts open cleanly and instantly.
- **Sharpening the Rate of Collapse ($\frac{di}{dt}$):** By absorbing the breaking current instantly, the condenser forces the primary current to drop to zero in a fraction of a millisecond. This creates an extreme $\frac{di}{dt}$ value, which maximizes the primary voltage spike and drastically increases the secondary induced voltage transferred to the output spark gap.
- **Resonant LC Ringing:** Once the condenser absorbs the charge, it momentarily bounces the energy back into the primary coil in a brief, high-frequency oscillatory ring (forming an LC tank

circuit with the primary winding). This rapid collapse and reversal of the magnetic field accelerates the induction process before the spring pulls the contacts fully back together.

User prompt: In the neef's hammer device the spark gap is connected to the secondary coil. Is that to amplify the voltage from the induction spike?

Response: Yes, the spark gap is connected to the secondary coil precisely to harness a massive, transformed voltage spike via mutual induction.

- **Turns Ratio Transformation:** While the primary circuit handles the low-voltage battery power and the high-current breaking spike, the secondary winding consists of thousands of turns of fine insulated wire. According to the transformer turns ratio ($V_{\text{sec}} = V_{\text{pri}} \times \frac{N_{\text{sec}}}{N_{\text{pri}}}$), that enormous primary voltage spike is multiplied by the massive ratio of secondary-to-primary turns.
 - **Stepping Up to Kilovolts:** Even though the primary inductive kick might only reach a few hundred volts, stepping that up through thousands of secondary turns pushes the potential to tens or hundreds of thousands of volts.
 - **Bridging the Gap:** This extreme secondary voltage is what allows the spark to violently ionize the air and jump across a wide external spark gap, turning a low-voltage DC current into a high-voltage, high-frequency discharge.
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User prompt: Do the turns in the secondary have to be in a single layer?

Response: No, the secondary winding in a classic induction coil is deliberately wound in **multiple layers** rather than a single layer.

- **Achieving Massive Turns Ratios:** To step a low battery voltage up to tens or hundreds of thousands of volts, the secondary requires thousands—and sometimes miles—of fine wire turns, which would be physically impossible to fit in a single layer along a core.
 - **The High-Voltage Breakdown Risk:** Stacking multiple layers introduces a severe design challenge because high voltages accumulate rapidly between adjacent layers. If a wire carrying thousands of volts ends up lying directly against a wire from a lower layer, the extreme potential difference will cause a spark to punch right through the enamel insulation and ruin the coil.
 - **Sectional Winding Solutions:** To prevent internal short circuits, historical coil builders like Heinrich Ruhmkorff divided the secondary spool into dozens of isolated, thin compartments or disk-like sections. This kept the cumulative voltage difference between adjacent wire turns low enough that paraffin-wax or resin insulation could safely contain it.
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User prompt: What about modern enameled wire?

Response: Modern magnet wire completely transforms the construction of induction coils by replacing bulky historical insulations with high-performance synthetic polymer coatings.

- **Ultra-Thin Dielectric Layers:** Modern enamels (such as modified polyurethane, polyester, or polyimide) are applied as microscopic films that offer immense dielectric breakdown strength—often several thousand volts per mil—drastically outperforming old silk, cotton, or paper wraps.
 - **Higher Fill Factor:** Because the polymer coating takes up virtually zero physical space compared to textile wrappings, you can pack significantly more copper turns into the same volume, yielding a much higher magnetic field and turns ratio.
 - **Thermal Resilience:** Modern magnet wire is rated for high thermal classes (ranging from Class B 130°C up to Class H 200°C or higher for polyimide/Kapton coats), meaning it can handle internal heat from high currents without the insulation scorching, melting, or degrading.
 - **Simplified Construction:** It eliminates the messy historical requirement to hand-impregnate every single layer with melted shellac, pitch, or paraffin wax during winding, though interleaving a thin modern plastic film (like Mylar or capacitor film) between deep secondary layers is still wise for extreme high-voltage builds to stop edge-to-edge tracking.
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User prompt: Ehat tools did faraday use for n his experiments?

Response: Michael Faraday relied on a mix of standard 19th-century laboratory apparatus, chemical batteries, and remarkably humble everyday materials to build his breakthrough devices:

- **Voltaic Cells and Batteries:** He used early chemical batteries (like the voltaic piles and trough configurations given to him by Alessandro Volta and developed by Humphry Davy) to supply steady DC currents for his electromagnetic experiments.
 - **Mercury Baths:** For his early work on electromagnetic rotation (the first primitive electric motor), he used glass vessels partially filled with liquid mercury to act as a fluid, continuous electrical conductor.
 - **Improvised Wire and Insulation:** When constructing his famous 1831 induction ring (the first transformer), he famously wound coils using ordinary copper wire intended for ladies' bonnets, insulating the turns by hand with strips of cotton cloth.
 - **Iron Rings and Soft-Iron Cores:** He wound coils around specially forged soft-iron rings and bar magnets to concentrate magnetic flux and study how magnetic fields interact with electric circuits.
 - **Galvanometers:** To detect and measure the tiny, fleeting electric currents produced by induction, he utilized sensitive early galvanometers—compass needles wrapped in multi-turn wire coils.
 - **Heavy Electromagnets and Optical Glass:** In his later research on diamagnetism and the magneto-optical effect, he employed a massive custom iron-core electromagnet powered by large battery banks, alongside blocks of heavy, highly dense optical glass that he had formulated himself years earlier.
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